

Chapter 17 Magnetic Field

17.1 Magnetic field

It is observed as early as centuries before Christ, that certain iron ores, such as lodestone, have the property of attracting small pieces of irons.

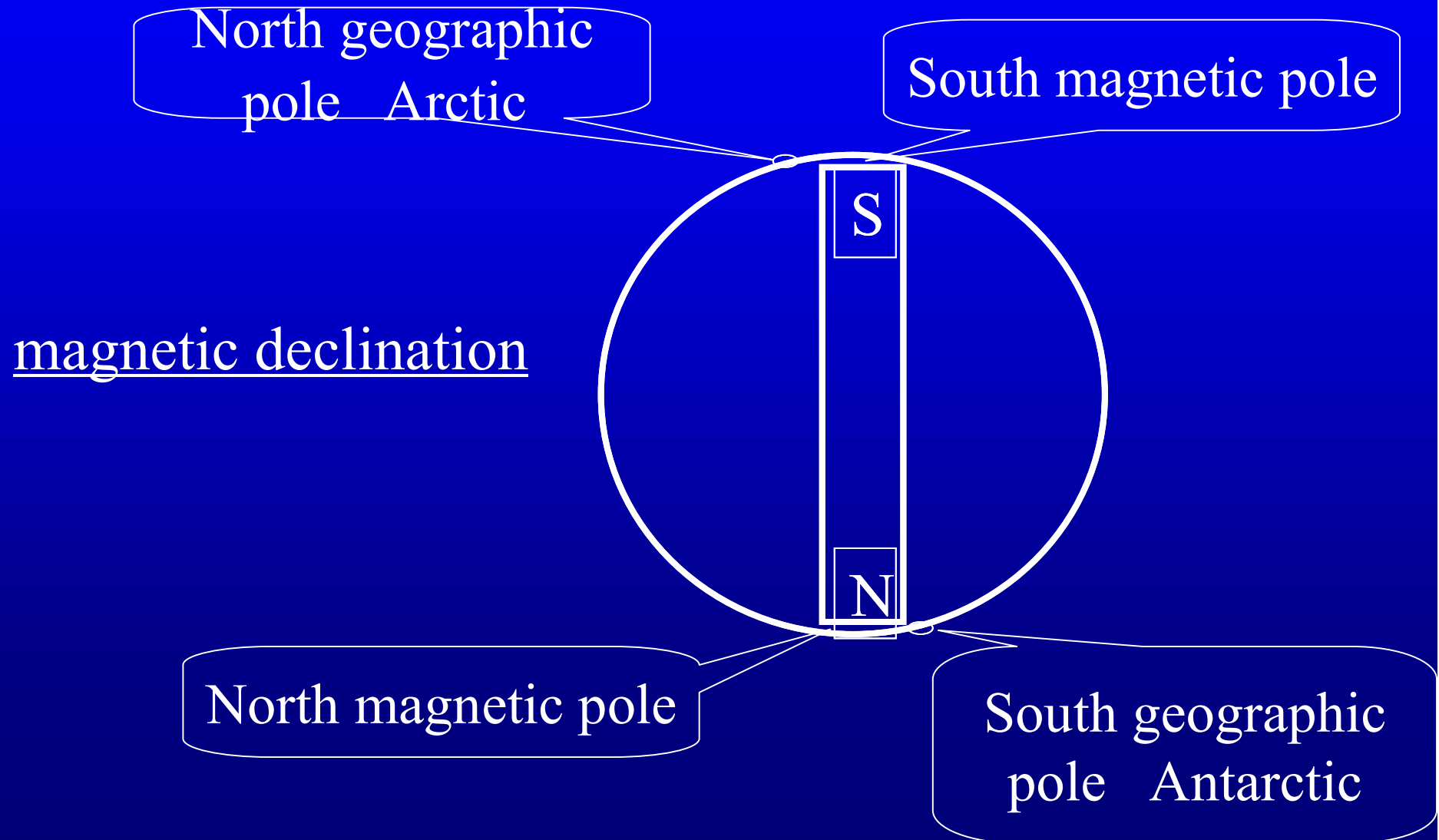
iron, cobalt and manganese and many compounds of them.

magnetism magnet

The magnetism seems to be concentrated on two regions of the magnet---magnetic poles.

N (the north seeking) pole

S (the south seeking) pole



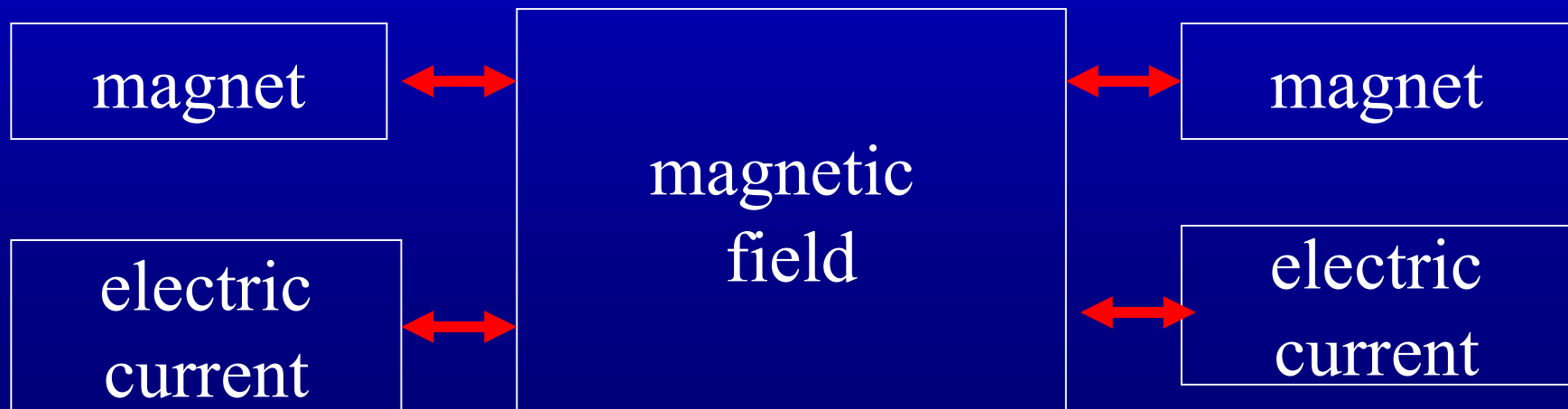
like magnetic poles repulsion

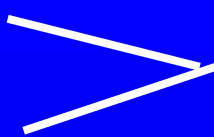
unlike magnetic poles attraction

1819 Oersted “He was quite struck with perplexity to see the needle making a great oscillation”

magnet-magnet, electric current-magnet and current-current

an interaction at a distance

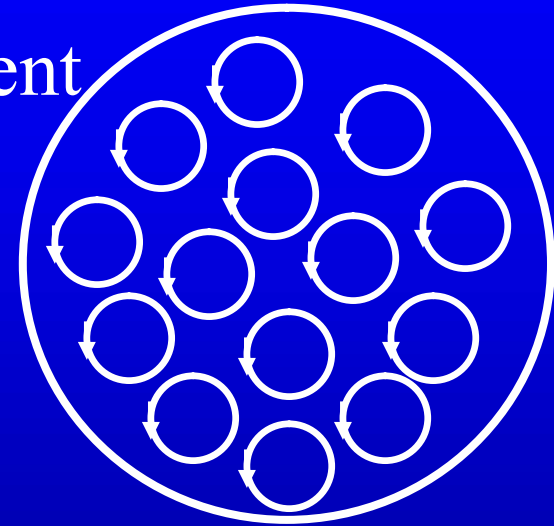


magnetic rod
solenoid  similar fields

Ampere's molecular electronic current

Electronic orbits with definite
angular momentum.

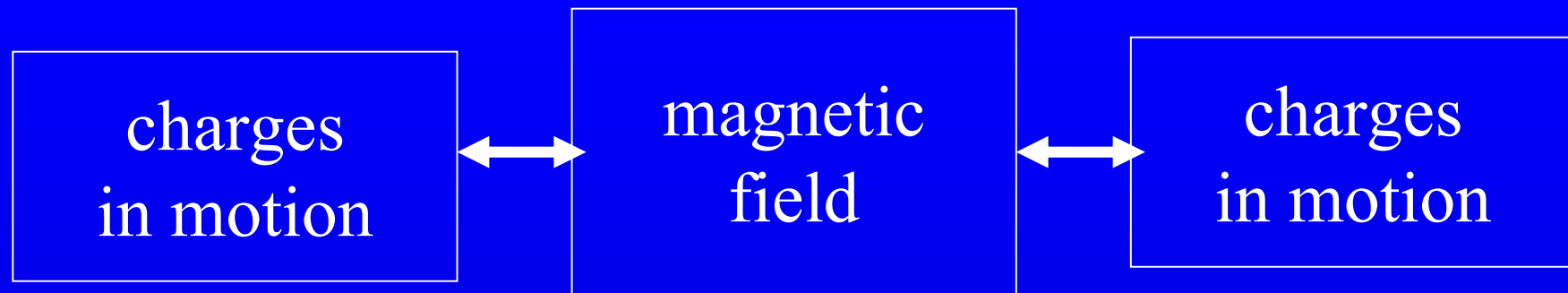
The magnetism is a
manifestation of electric
charge in motion.



internal currents
cancel each other



net current on
the surface



The definition of magnetic field

electric field

test charge q_0

$$\mathbf{E} = \frac{\mathbf{F}}{q_0}.$$

magnetic field

$v=0 \longrightarrow F=0$

moving test charge q_0

(1) The force acted by a magnetic field on a moving charge is proportional to the velocity of the charge and its direction is perpendicular to the velocity.

(2) The force acted on the moving charge is zero when it moves along a certain direction. This direction is defined as the definition of the magnetic field (or magnetic induction (intensity)) $\mathbf{B}$.

(3) When $\mathbf{v}$ is perpendicular to $\mathbf{B}$, the force F takes its maximum value $F_{\max}$. Then the magnitude of $\mathbf{B}$ is defined as $B = F_{\max} / q_0 v$.

magnetic force per unit $q_0 v$

$$\mathbf{F} = q_0 \mathbf{v} \times \mathbf{B}$$

unit: $\text{N}/(\text{A} \cdot \text{m})$ tesla (T)

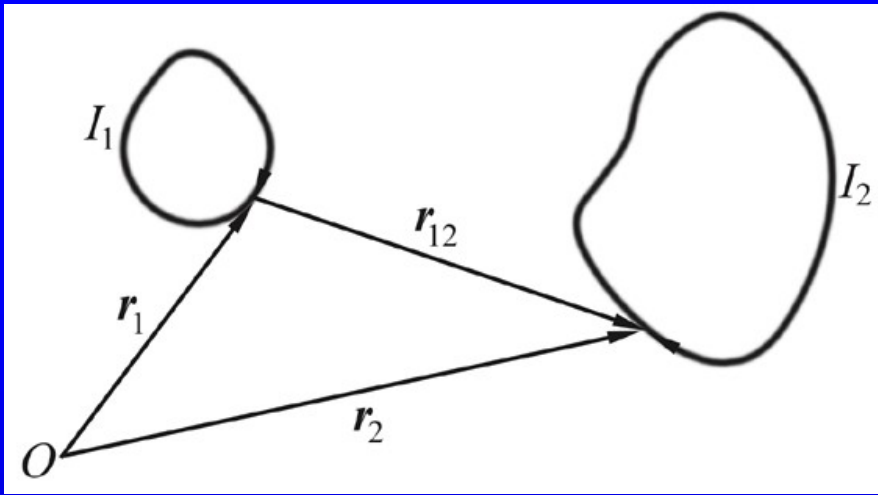
$$\mathbf{F} = q_0(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Lorentz force

For a current element in a magnetic field

$$d\mathbf{F} = I d\mathbf{l} \times \mathbf{B}.$$

On October 1820, French scientists Jean-Baptiste Biot and Félix Savart proposed the formula for the magnetic field around a current, while on December 1820 **Ampère** brought forward an expression for interaction between two closed current loops.



the force acted on
loop2 by loop1.

$$\mathbf{F}_{21} = \frac{\mu_0}{4\pi} \oint_{l_2} \oint_{l_1} \frac{I_2 d\mathbf{l}_2 \times (I_1 d\mathbf{l}_1 \times \mathbf{r}_{12})}{r_{12}^3}.$$

magnetic permeability
of free space

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{N}}{\text{A}^2}$$

The force acted on a current element $I_2 d\mathbf{l}_2$ on loop2 is

$$\mathbf{F}_{21} = I_2 d\mathbf{l}_2 \times \frac{\mu_0}{4\pi} \oint_{l_1} \frac{I_1 d\mathbf{l}_1 \times \mathbf{r}_{12}}{r_{12}^3} = I_2 d\mathbf{l}_2 \times \mathbf{B}_2,$$

where $\mathbf{B}_2 = \frac{\mu_0}{4\pi} \oint_{l_1} \frac{I_1 d\mathbf{l}_1 \times \mathbf{r}_{12}}{r_{12}^3}$ is the magnetic field produced by loop 1 at the current element $I_2 d\mathbf{l}_2$.

$$\mathbf{F}_{21} = \oint_{l_2} I_2 d\mathbf{l}_2 \times \mathbf{B}_2.$$

Generally, the field produced by a closed current loop is

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{I d\mathbf{l}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad \text{Biot-Savart's law}$$

The field produced by a current element is

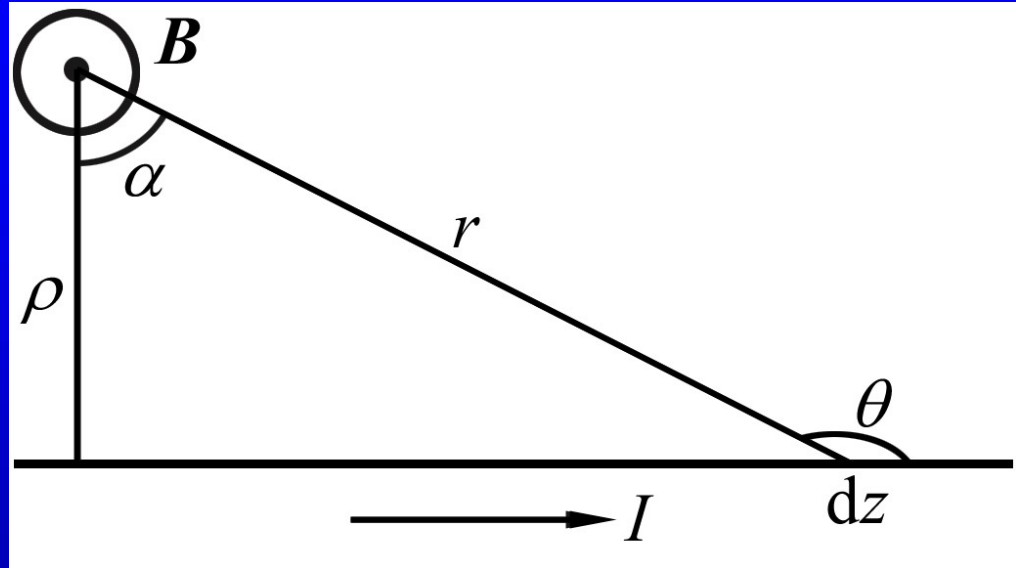
$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \mathbf{r}}{r^3}$$

Example 17.1 The steady magnetic field near a long straight current-carrying wire.

Solution: The magnetic field produced by the current element $I dz$ is

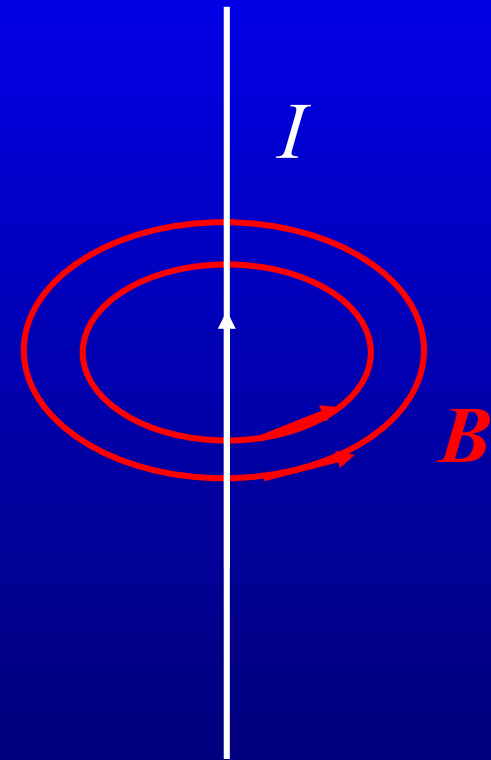
$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I \mathbf{e}_z dz \times \mathbf{r}}{r^3}$$

$$= \frac{\mu_0}{4\pi} \frac{I \mathbf{e}_z dz \times (\rho \mathbf{e}_\rho - z \mathbf{e}_z)}{(\sqrt{z^2 + \rho^2})^3} = \frac{\mu_0}{4\pi} \frac{I \rho dz}{(\sqrt{z^2 + \rho^2})^3} \mathbf{e}_\phi.$$



Therefore, the resultant magnetic field is

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0 I \rho}{4\pi} \mathbf{e}_\varphi \int_{-\infty}^{\infty} \frac{dz}{(\sqrt{z^2 + \rho^2})^3} \\ &= \frac{\mu_0 I \rho}{4\pi} \mathbf{e}_\varphi \left[\frac{z}{\rho^2 \sqrt{z^2 + \rho^2}} \right]_{-\infty}^{+\infty} \\ &= \frac{\mu_0 I}{2\pi \rho} \mathbf{e}_\varphi. \end{aligned}$$



$$dB = \frac{\mu_0}{4\pi} \frac{Idz r \sin \theta}{r^3}$$

$$\sin \theta = \cos \alpha,$$

$$r = \frac{\rho}{\cos \alpha} = \rho \sec \alpha$$

$$z = \rho \tan \alpha, dz = \rho \sec^2 \alpha d\alpha.$$

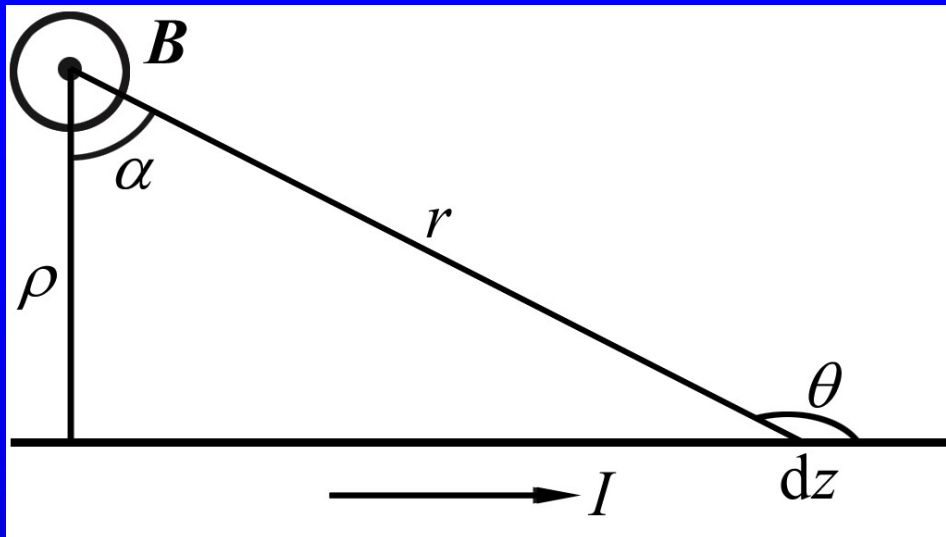
$$|Idz \mathbf{e}_z \times \mathbf{r}| = Idz r \sin \theta$$

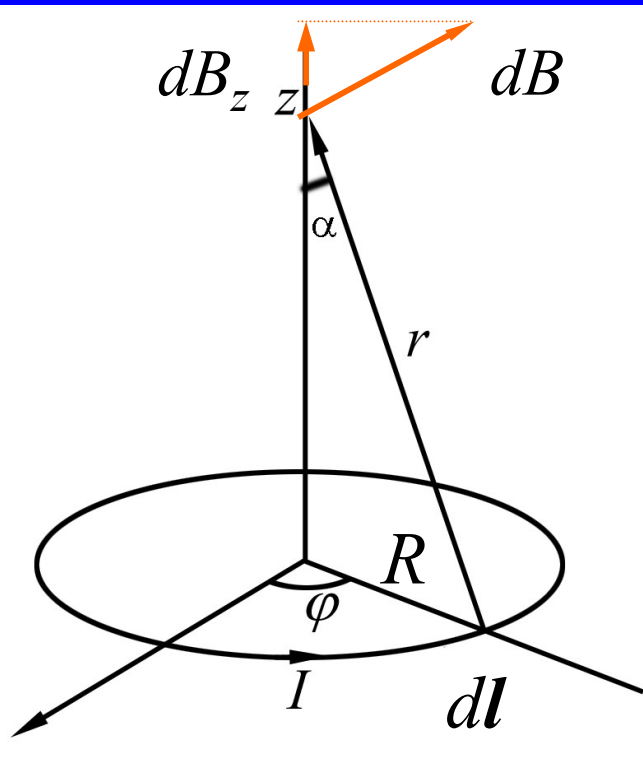
$$dB = \frac{\mu_0}{4\pi} \frac{Idz \cos \alpha}{r^2} = \frac{\mu_0}{4\pi} \frac{I \rho \sec^2 \alpha \cos \alpha d\alpha}{\rho^2 \sec^2 \alpha}$$

$$= \frac{\mu_0 I}{4\pi} \frac{\cos \alpha d\alpha}{\rho}$$

$$B = \frac{\mu_0 I}{4\pi \rho} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \alpha d\alpha = \frac{\mu_0 I}{2\pi \rho}.$$

secant cosecant





Example 17.3 Magnetic field along the axis of a current-carrying loop.

Solution: We see in the figure that r and dl are perpendicular. So, according to Biot-Savart law,

$$dB = \frac{\mu_0}{4\pi} \frac{Idl}{r^2}.$$

The component along the axis is

$$dB_z = \frac{\mu_0}{4\pi} \frac{Idl}{r^2} \sin \alpha = \frac{\mu_0 I}{4\pi} \frac{R}{r^3} dl.$$

Integration around the loop gives

$$B_z = \frac{\mu_0 I}{4\pi} \frac{R}{r^3} (2\pi R) = \frac{\mu_0 I}{2} \frac{R^2}{(z^2 + R^2)^{3/2}}.$$

The magnetic dipole moment can be defined as

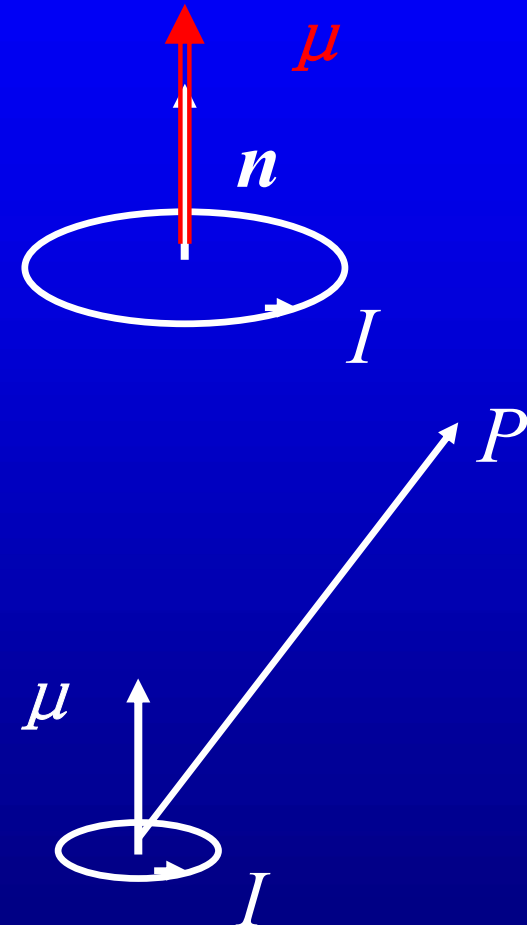
$$\mu = IS = \pi R^2 In.$$

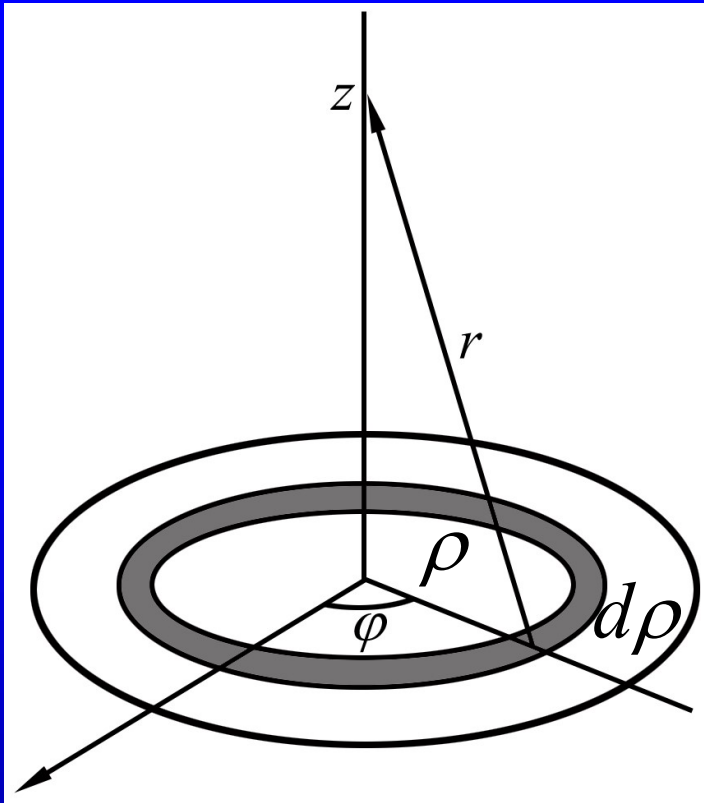
When $z \gg R$, we have

$$B = \frac{\mu_0}{4\pi} \frac{2\mu}{(z^2 + R^2)^{3/2}} \underset{z \gg R}{\approx} \frac{\mu_0}{4\pi} \frac{2\mu}{z^3}.$$

The off-axis magnetic field of a magnetic dipole is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\boldsymbol{\mu} \cdot \mathbf{e}_r) \mathbf{e}_r - \boldsymbol{\mu}].$$





Example 17.5 Find the magnetic field along the axis of a rotating charged disc with surface density σ .

Solution: Divide the disc into many rings as show in the figure. The electric current on a ring due to the rotation of the disc is

$$dI = \sigma \rho \omega d\phi.$$

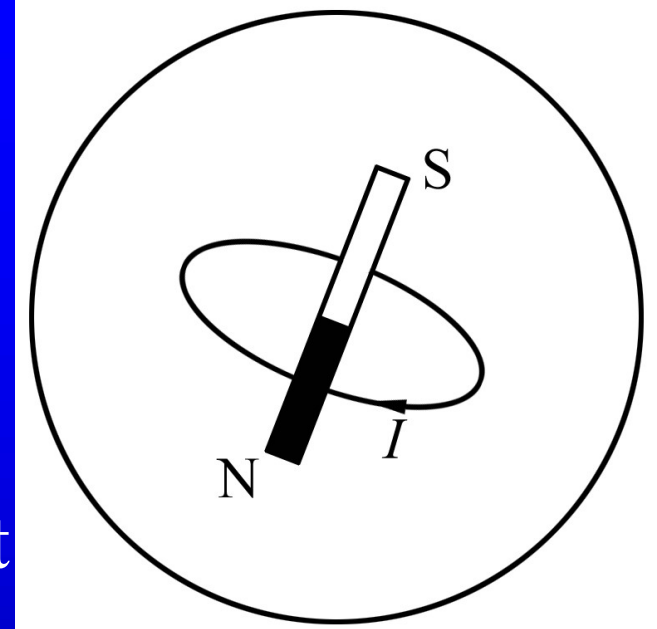
Then, the magnetic field produced by the current on the ring is

$$dB_z = \frac{\mu_0}{2} \frac{\rho^2 dI}{(z^2 + \rho^2)^{3/2}} = \frac{\mu_0 \sigma \omega}{2} \frac{\rho^3 d\rho}{(z^2 + \rho^2)^{3/2}}$$

The resultant magnetic field is

$$\begin{aligned} B_z &= \int dB_z = \frac{\mu_0 \sigma \omega}{2} \int_0^R \frac{\rho^3 d\rho}{(z^2 + \rho^2)^{3/2}} \\ &= \frac{\mu_0 \sigma \omega}{2} \left[\frac{2z^2 + R^2}{(z^2 + R^2)^{1/2}} - 2|z| \right]. \end{aligned}$$

Example 17.6 Suppose the earth's magnetic field is produced by a current circling around underneath the equator at an average distance of 5000km from the center. What would be the value of it if the field at either pole is about 10^{-4}T (1 gauss).



Solution: From Example 17.3, the field at magnetic pole is determined by

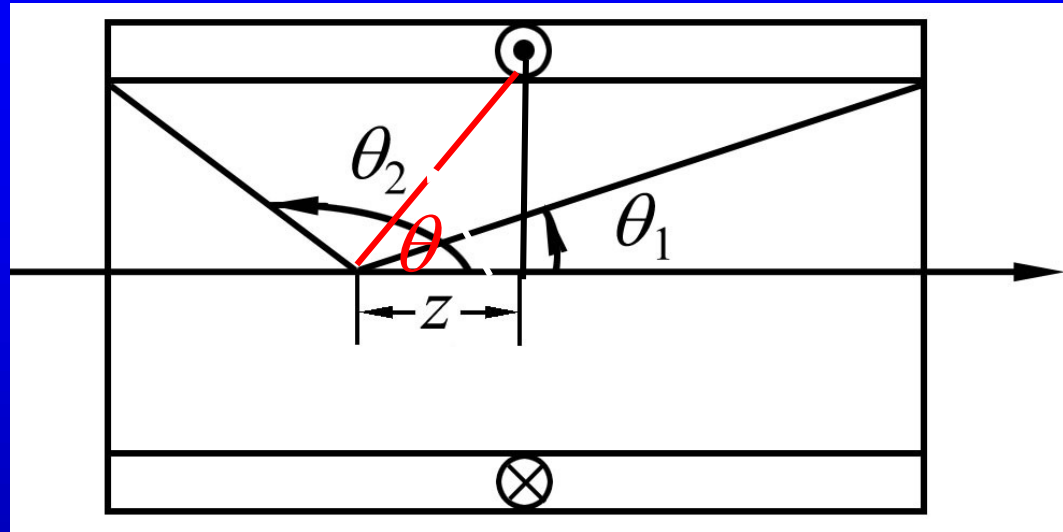
$$B = \frac{\mu_0}{4\pi} \frac{2\pi IR^2}{r^3},$$

where $R = 5000$ km and $r = 8100$ km is the distance from the current to magnetic pole.

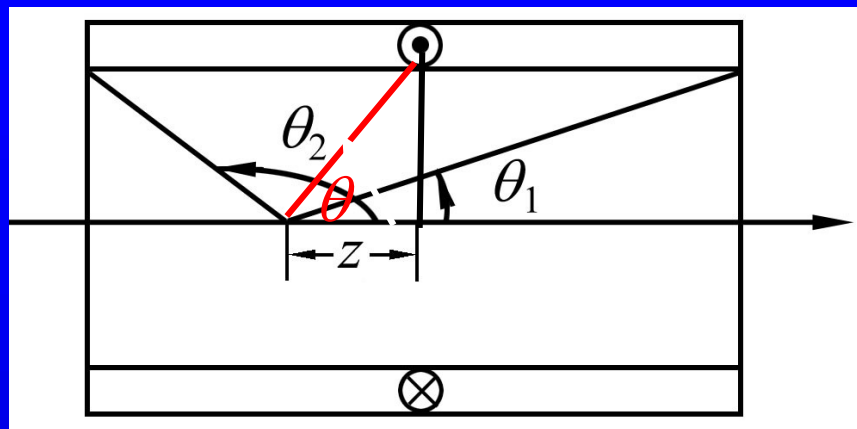
Therefore, the current is estimated to be

$$I = \frac{Br^3}{\frac{\mu_0}{4\pi} 2\pi R^2} \approx 3.4 \times 10^9 \text{ A.}$$

Example 17.7 Find the magnetic field along the axis of a solenoid.



Solution: As shown in the figure, a solenoid is composed of several coaxial circular loops of the same radius that carry the same current. Assuming the number of turns per unit length is n , the field produced by loops in the section dz is



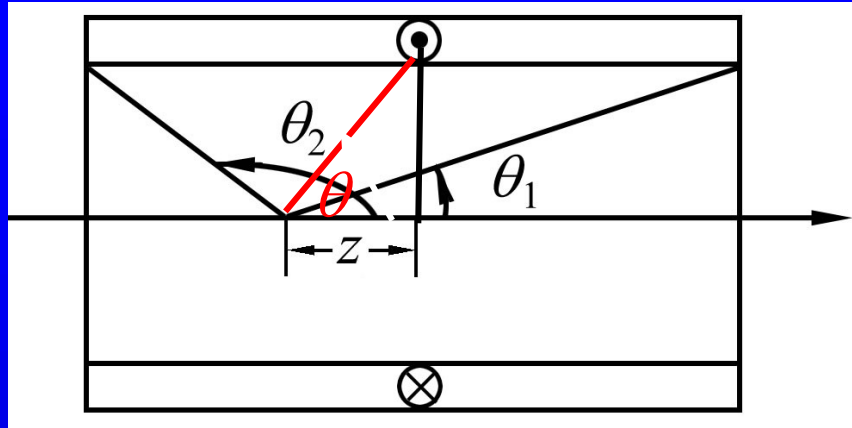
$$dB_z = \frac{\mu_0}{2} \frac{I(ndz)R^2}{(z^2 + R^2)^{3/2}} = \frac{\mu_0 nI}{2} \frac{R^2 dz}{(z^2 + R^2)^{3/2}}$$

From the figure, $z = R \cot \theta$, then

$$dz = -R \csc^2 \theta d\theta, \quad z^2 + R^2 = R^2 \csc^2 \theta,$$

and the above equation become

$$dB_z = \frac{\mu_0 nI}{2} (-\sin \theta) d\theta.$$



The resultant field can be obtained by integrating from one end of the solenoid to the other, i. e. ,

$$B_z = \frac{\mu_0 n I}{2} \int_{\theta_2}^{\theta_1} -\sin \theta d\theta = \frac{\mu_0 n I}{2} (\cos \theta_1 - \cos \theta_2).$$

For an infinite solenoid, $\theta_1 = 0$ and $\theta_2 = \pi$ respectively. The result is $B_z = \mu_0 n I$.

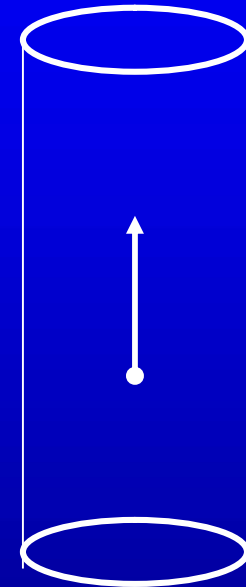
Example Find the (nonrelativistic) magnetic field of a moving point charge.

$$\mathbf{B} = \frac{\mu_0}{4\pi} \oint \frac{I d\mathbf{l} \times \mathbf{r}}{r^3}.$$

$$I d\mathbf{l} = j S d\mathbf{l} = j S d\mathbf{l} \mathbf{e}_T = j d\tau,$$

$$\mathbf{j} = nq\mathbf{v}.$$

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{(nq\mathbf{v}) \times \mathbf{r}}{r^3} d\tau = \frac{\mu_0}{4\pi} \int \frac{q\mathbf{v} \times \mathbf{r}}{r^3} n d\tau.$$



Each charge produces a magnetic field given by

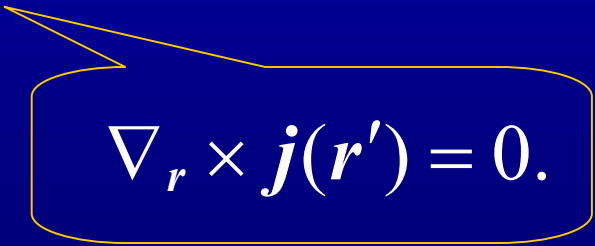
$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3}.$$

*The vector potential

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \oint \frac{I d\mathbf{l}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} = \frac{\mu_0}{4\pi} \oint I d\mathbf{l}' \times \left(-\nabla_r \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \\ &= \frac{\mu_0}{4\pi} \oint \left(\nabla_r \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \times I d\mathbf{l}' \end{aligned}$$

$$I d\mathbf{l}' \rightarrow \mathbf{j}(\mathbf{r}') d\tau', \quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \left(\nabla_r \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \times \mathbf{j}(\mathbf{r}') d\tau'$$

Using $\nabla \times (\varphi \mathbf{f}) = \varphi(\nabla \times \mathbf{f}) + (\nabla \varphi) \times \mathbf{f}$


$$\nabla_r \times \mathbf{j}(\mathbf{r}') = 0.$$

$$\mathbf{B}(\mathbf{r}) = \nabla_{\mathbf{r}} \times \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{j}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\tau'$$

$$\equiv \nabla \times \mathbf{A}(\mathbf{r})$$

Vector potential

17.2 Magnetic Gauss' law and Ampère's circuital law

The magnetic flux is defined as

$$\Phi_m = \int_S \mathbf{B} \cdot d\mathbf{S}.$$

Unit: $\text{T}\cdot\text{m}^2$ weber (Wb)

The **magnetic Gauss' law** reads

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

The lines of force of magnetic field is closed.

Using Gauss' theorem, we have

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{B} d\tau = 0. \quad \Rightarrow \quad \nabla \cdot \mathbf{B} = 0.$$

There are no source for the magnetic field.

There exists no magnetic charge or magnetic monopole.

For the magnetic field of a steady current

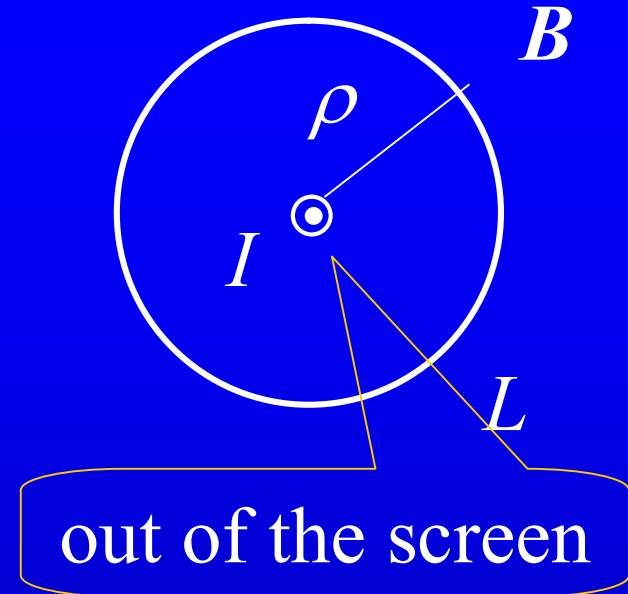
$$\mathbf{B} = \nabla \times \mathbf{A}. \quad \Rightarrow \quad \nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

Ampère's circuital law

L — a circular path surrounding I

$$\mathbf{B} = \frac{\mu_0 I}{2\pi\rho} \mathbf{e}_\phi.$$

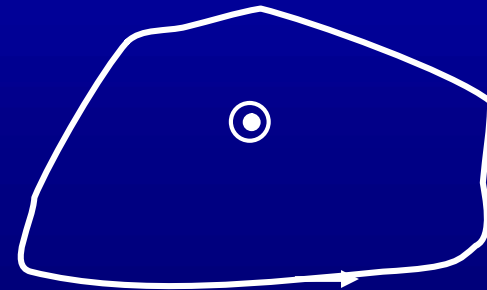
$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi\rho} \cdot 2\pi\rho = \mu_0 I.$$



L — an arbitrary path surrounding I

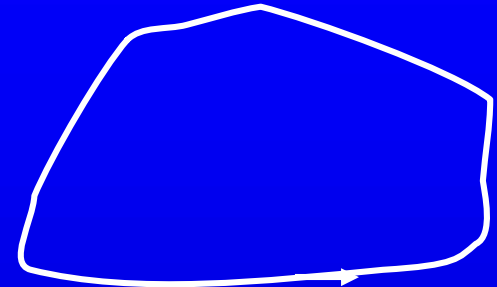
$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \oint_L \frac{\mu_0 I}{2\pi\rho} \mathbf{e}_\phi \cdot d\mathbf{l}$$

$$= \frac{\mu_0 I}{2\pi} \oint_L \frac{\mathbf{e}_\phi \cdot d\mathbf{l}}{\rho} = \frac{\mu_0 I}{2\pi} \oint_L d\phi = \mu_0 I.$$



If I does not pass through L , we have

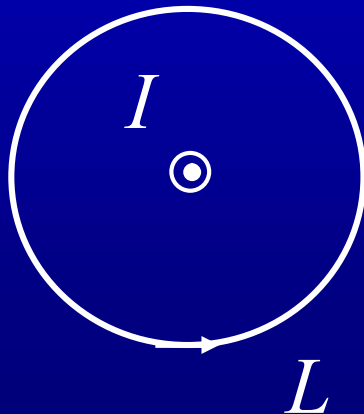
$$\oint_L \mathbf{B} \cdot d\mathbf{l} = 0.$$



These relations are also satisfied by all steady currents moving in curves as well as in straight lines.

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \mu_0 \sum_{\text{inside } L} I.$$

Ampère's circuital law



positive
current



negative
current

right-hand
rule

***Ampère's circuital law in differential form**

Ampère's law can be rewritten as

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_S \mathbf{j} \cdot d\mathbf{S} = \mu_0 I,$$

where S is a surface that its edge is L . Using Stokes' theorem for the left side of above equation, we have

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{S}.$$

Therefore, we finally obtain

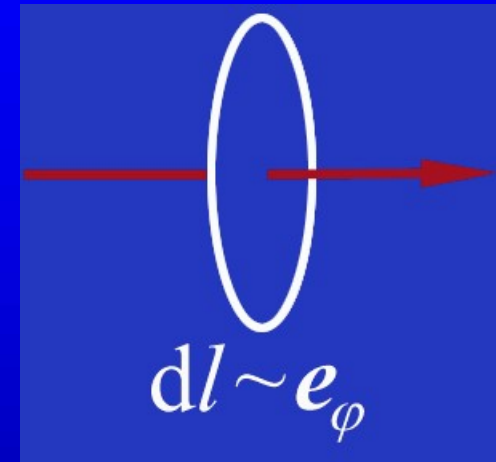
$$\text{rot } \mathbf{B} = \nabla \times \mathbf{B} = \mu_0 \mathbf{j}.$$

That is Ampère's circuital law in differential form.

Example 17.8 Using Ampere's law, find the magnetic field of a rectangular current.

$$\oint \mathbf{B} \cdot d\mathbf{l} = B_{\phi} 2\pi \rho = \mu_0 I.$$

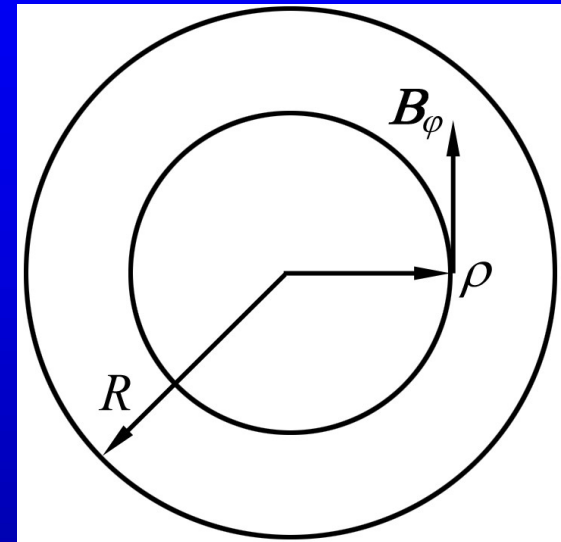
$$B_{\phi} = \frac{\mu_0 I}{2\pi \rho}.$$



Example 17.9 Determine the magnetic field of an **infinite cylinder** of uniform current.

Solution: The symmetry of the problem suggests that the lines of force of the field are circles concentric with the axis of the cylinder. Therefore, choosing a circle of radius ρ as the path L , the circulation of the magnetic field is

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi\rho.$$



For $\rho > R$, Ampere's law gives

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi\rho = \mu_0 I, \quad \text{or} \quad B = \frac{\mu_0 I}{2\pi\rho}.$$

For $\rho < R$, the current passing through L is

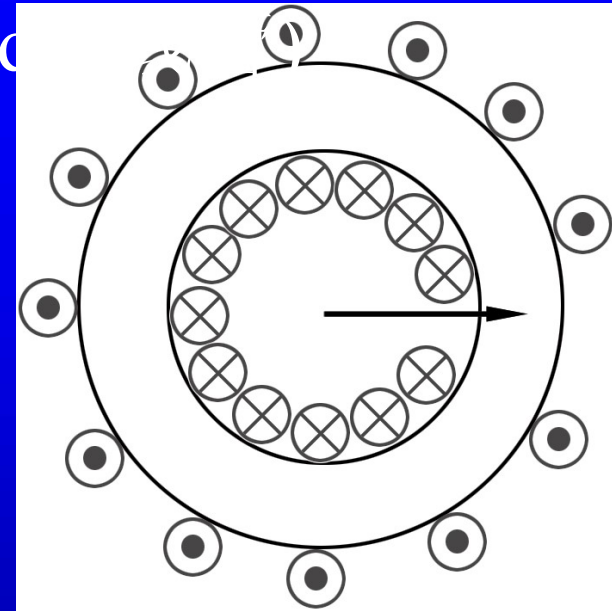
$$I' = \frac{I}{\pi R^2} \pi \rho^2 = \frac{I \rho^2}{R^2}.$$

Therefore, using Ampere's law, we have

$$B \cdot 2\pi\rho = \frac{\mu_0 I \rho^2}{R^2}, \quad \text{or} \quad B = \frac{\mu_0 I \rho}{2\pi R^2}.$$

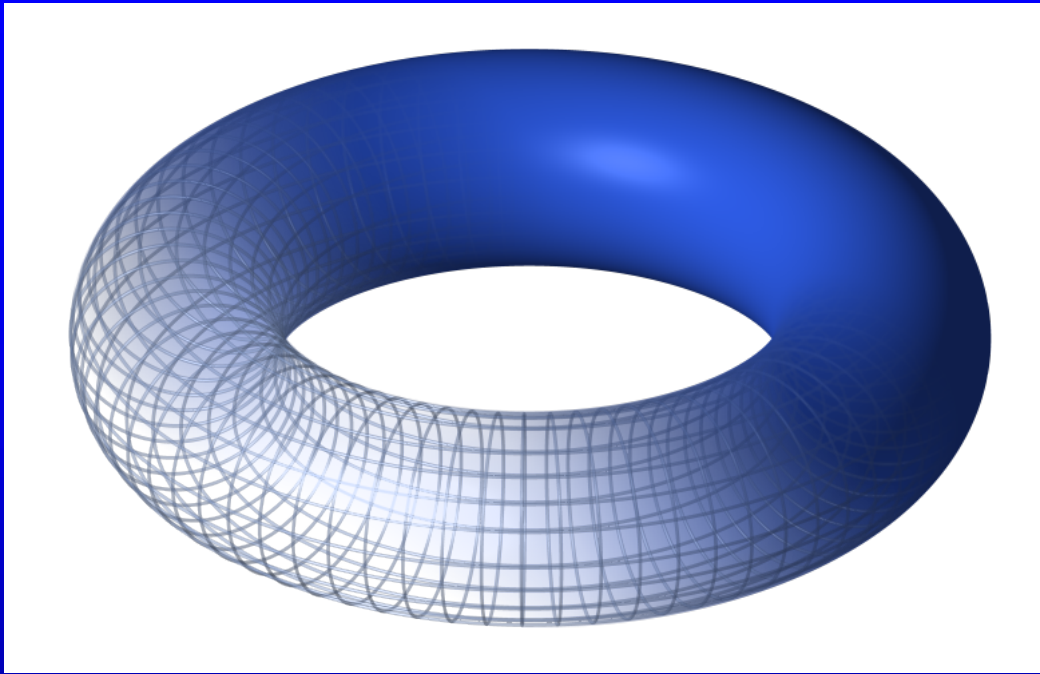
Example 17.11 Find the magnetic field of a toroid (toroidal coil)(螺绕环). (solenoid)

Solution: The symmetry of the problem suggests that the lines of force of the magnetic field are circles concentric with the torus.



Taking a path L that links all turns around the torus, the total current through is circulation of the magnetic field is NI , where N is the total turns of the toroid. Thus, Ampere's law gives

$$2\pi rB = \mu_0 NI.$$



torus

the shape of a donut

Then, we have

$$B = \frac{\mu_0 N I}{2\pi r}$$

At limit of large radius, we have

$$\begin{aligned}\lim_{r \rightarrow \infty} B &= \lim_{r \rightarrow \infty} \mu_0 \frac{N}{2\pi r} I \\ &= \mu_0 n I.\end{aligned}$$

For any path outside the torus, the total current through it is zero. Therefore, $B=0$.

Assignment: 17.1,17.2,17.4 or 17.5

17.3 Magnetic force

The magnetic force on an electric current is

$$\mathbf{F} = \oint I d\mathbf{l} \times \mathbf{B}.$$

The force acted on a piece of wire of length l of one of two parallel conducting wire is

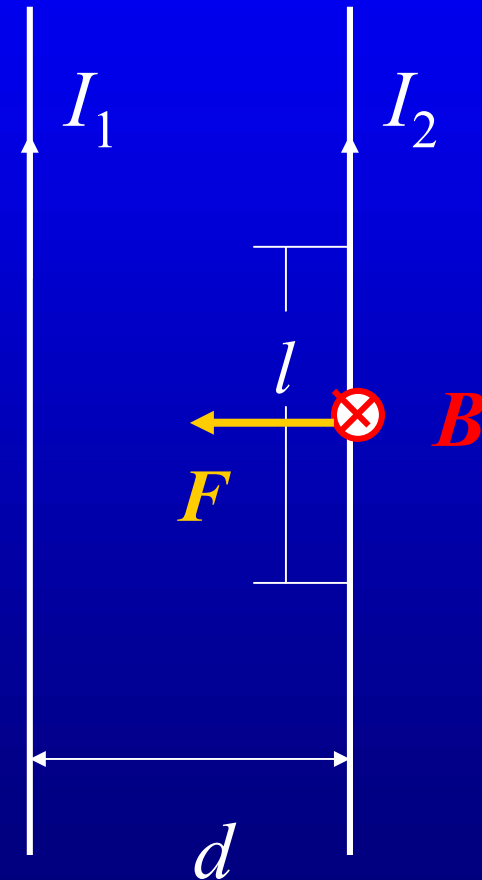
$$\mathbf{F} = I_2 d\mathbf{l} \times \mathbf{B}.$$

Since

$$B = \frac{\mu_0}{4\pi} \frac{2I_1}{d},$$

then

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} l,$$



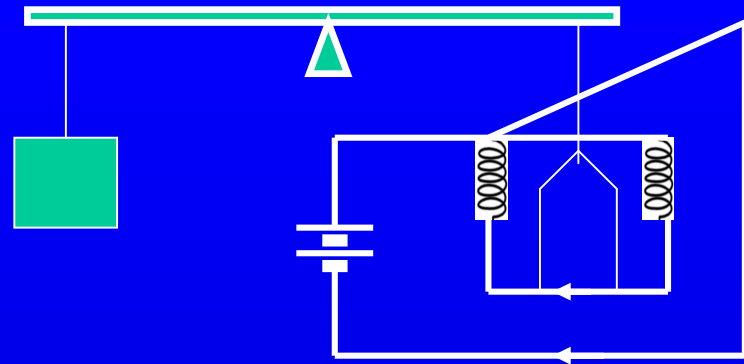
or $\frac{F}{l} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d}$.

$\frac{\mu_0}{4\pi}$ is assigned as 10^{-7}

If $d=1\text{m}$ and $\frac{F}{l} = 2 \times 10^{-7} \text{ N/m}$,

the current is defined as 1 ampère.

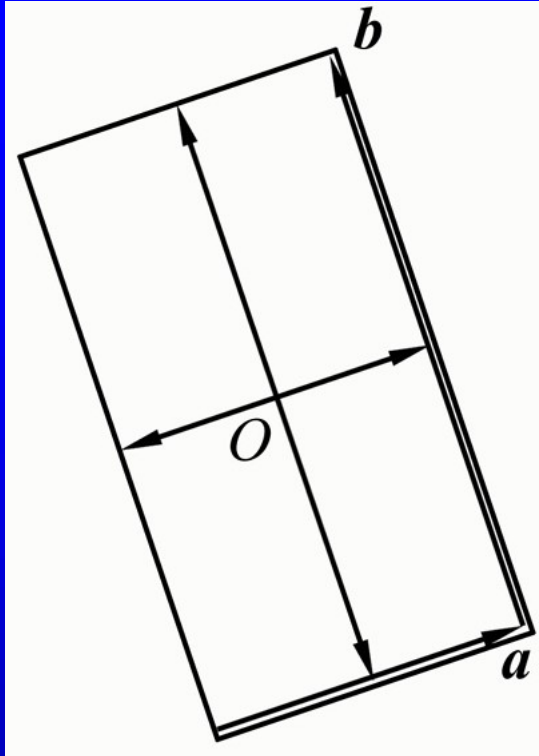
current balance



SI ----Système International 11th General Conference
on Weights and Measures ampère 4th fundamental unit

$$\frac{1}{\epsilon_0 \mu_0} = c^2 \Rightarrow \frac{1}{4\pi \epsilon_0} = 10^{-7} c^2$$

Practically, the current is easier to measure, but physically, the charge is more fundamental.



A loop of current in uniform field

$$\mathbf{F} = \oint I d\mathbf{l} \times \mathbf{B} = -\mathbf{B} \times I \oint d\mathbf{l} = \mathbf{0}$$

$$\begin{aligned} \mathbf{M} &= -\frac{\mathbf{b}}{2} \times (I\mathbf{a} \times \mathbf{B}) + \frac{\mathbf{a}}{2} \times (I\mathbf{b} \times \mathbf{B}) \\ &\quad + \frac{\mathbf{b}}{2} \times (-I\mathbf{a} \times \mathbf{B}) + \left(-\frac{\mathbf{a}}{2}\right) \times (-I\mathbf{b} \times \mathbf{B}) \\ &= -\mathbf{b} \times (I\mathbf{a} \times \mathbf{B}) + \mathbf{a} \times (I\mathbf{b} \times \mathbf{B}) \\ &= I(\mathbf{a} \times \mathbf{b}) \times \mathbf{B} = \boldsymbol{\mu} \times \mathbf{B} \end{aligned}$$

$$\mathbf{M} = \boldsymbol{\mu} \times \mathbf{B}$$

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} + C \quad (\text{C is constant})$$

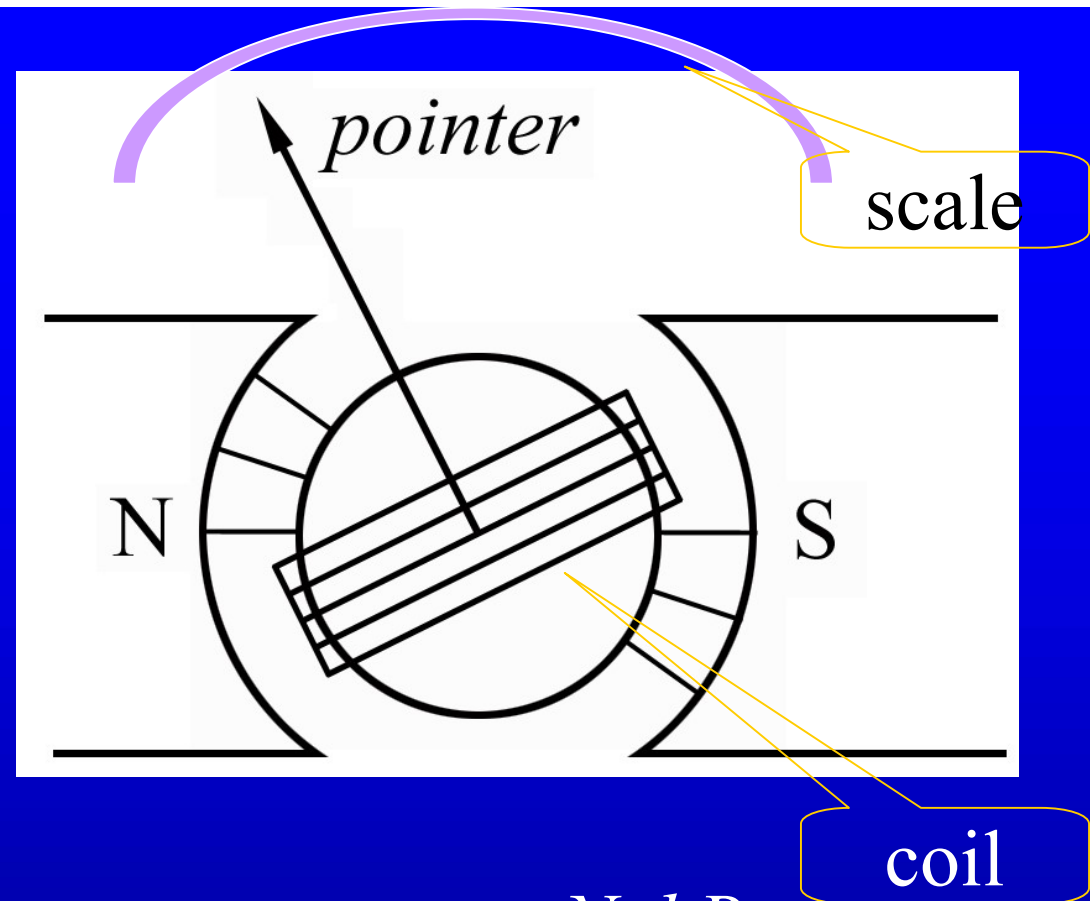
*Galvanometer

$$M = NI \cdot ab \cdot B$$

μ

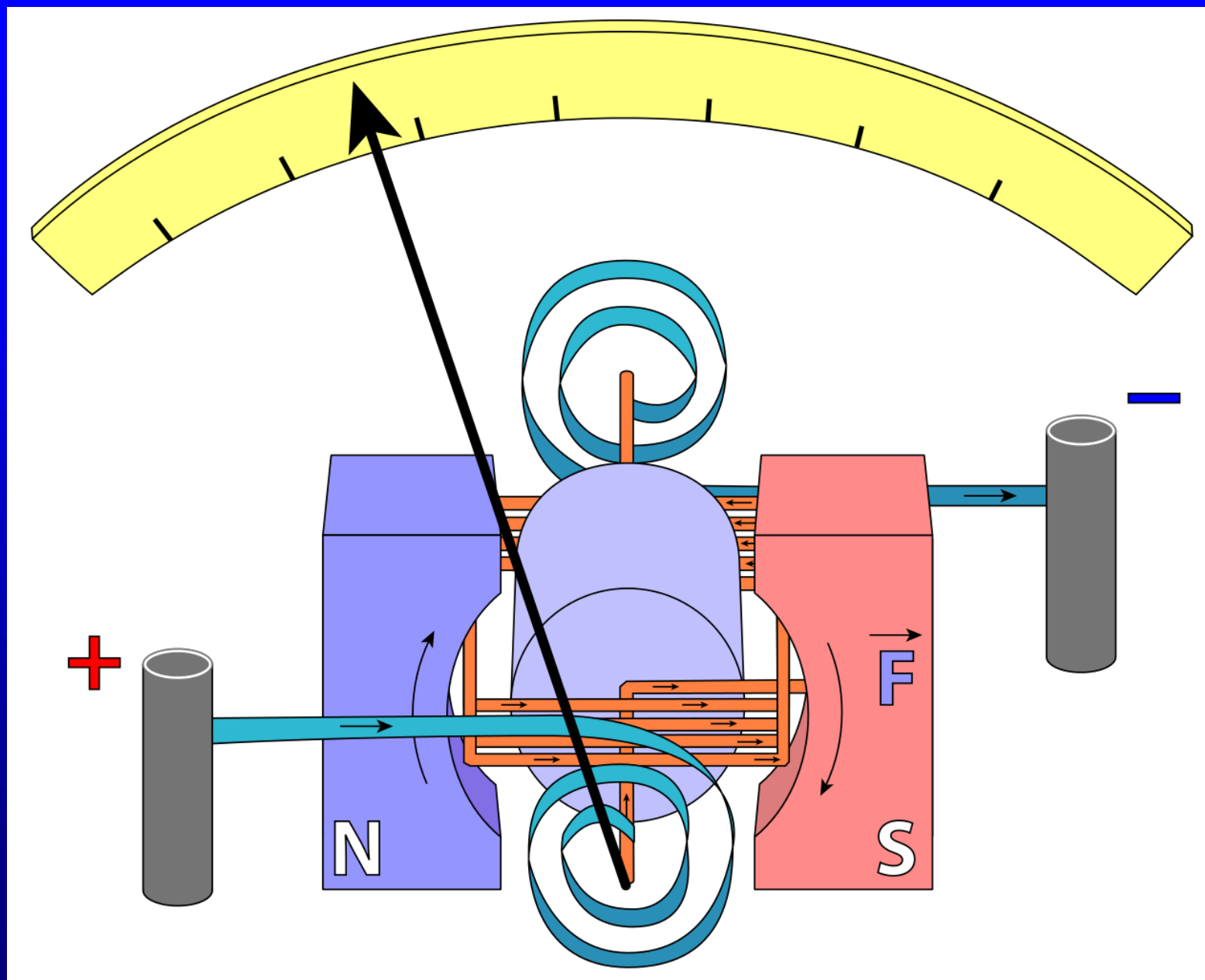
Torsion torque of
the spring

$$M' = \kappa \varphi$$



At equilibrium $M = M' \Rightarrow \varphi = \frac{NabB}{\kappa} I.$

The scale can be calibrated so that the value of the current can be read directly.



Motion of a charge in a magnetic field

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}.$$

$\therefore \mathbf{F} \perp \mathbf{v}$ $\therefore$ the magnetic field does not produce any change in kinetic energy of the particle.

If a charged particle is moving in a uniform magnetic field, the equation of motion is

$$m \frac{d\mathbf{v}}{dt} = q\mathbf{v} \times \mathbf{B}.$$

Assuming the magnetic field is in z-axis direction,
i. e.

$$\mathbf{B} = B\mathbf{e}_z.$$

the equation of motion can be rewritten as

$$\begin{cases} \frac{dv_x}{dt} = \frac{qB}{m} v_y \\ \frac{dv_y}{dt} = -\frac{qB}{m} v_x \\ \frac{dv_z}{dt} = 0 \end{cases}$$

The solution is

$$\begin{cases} v_x = v_0 \cos(\omega t + \delta) \\ v_y = -v_0 \operatorname{sgn}(q) \cos(\omega t + \delta) \\ v_z = C \end{cases}, \quad \omega = \frac{|q|B}{m}.$$

It is a circular motion ($v_z=0$) or helical motion ($v_z \neq 0$).

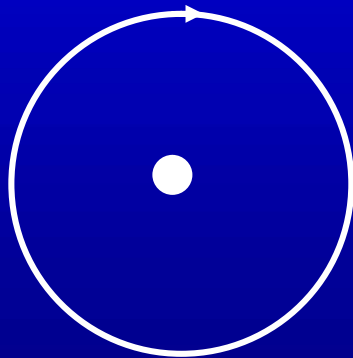
Suppose the angular velocity is ω . The centripetal acceleration is then

$$\mathbf{a} = \boldsymbol{\omega} \times \mathbf{v}.$$

The equation of motion becomes

$$m(\boldsymbol{\omega} \times \mathbf{v}) = q(\mathbf{v} \times \mathbf{B}) = -q\mathbf{B} \times \mathbf{v}.$$

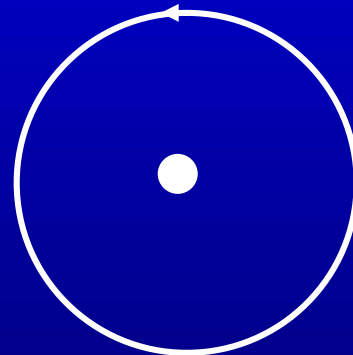
Thus $\boldsymbol{\omega} = -\frac{q}{m}\mathbf{B}.$



positive charge

$\mathbf{B}$ upward

$\boldsymbol{\omega}$ downward



negative charge

$\mathbf{B}$ upward

$\boldsymbol{\omega}$ upward

$$\omega = \frac{qB}{m} \quad \text{---cyclotron frequency}$$

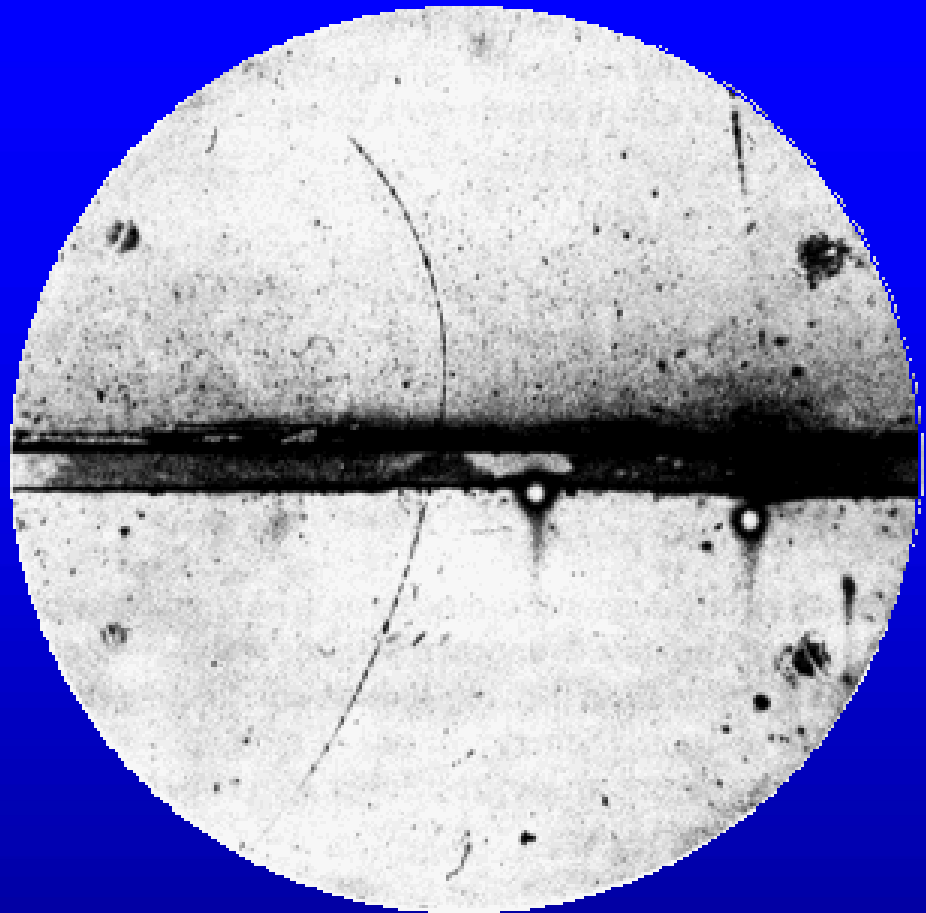
$$\rho = \frac{v_{\perp}}{\omega} = \frac{mv_{\perp}}{qB} \quad \text{---cyclotron radius}$$

Discovery of positron

Carl David Anderson,
1932

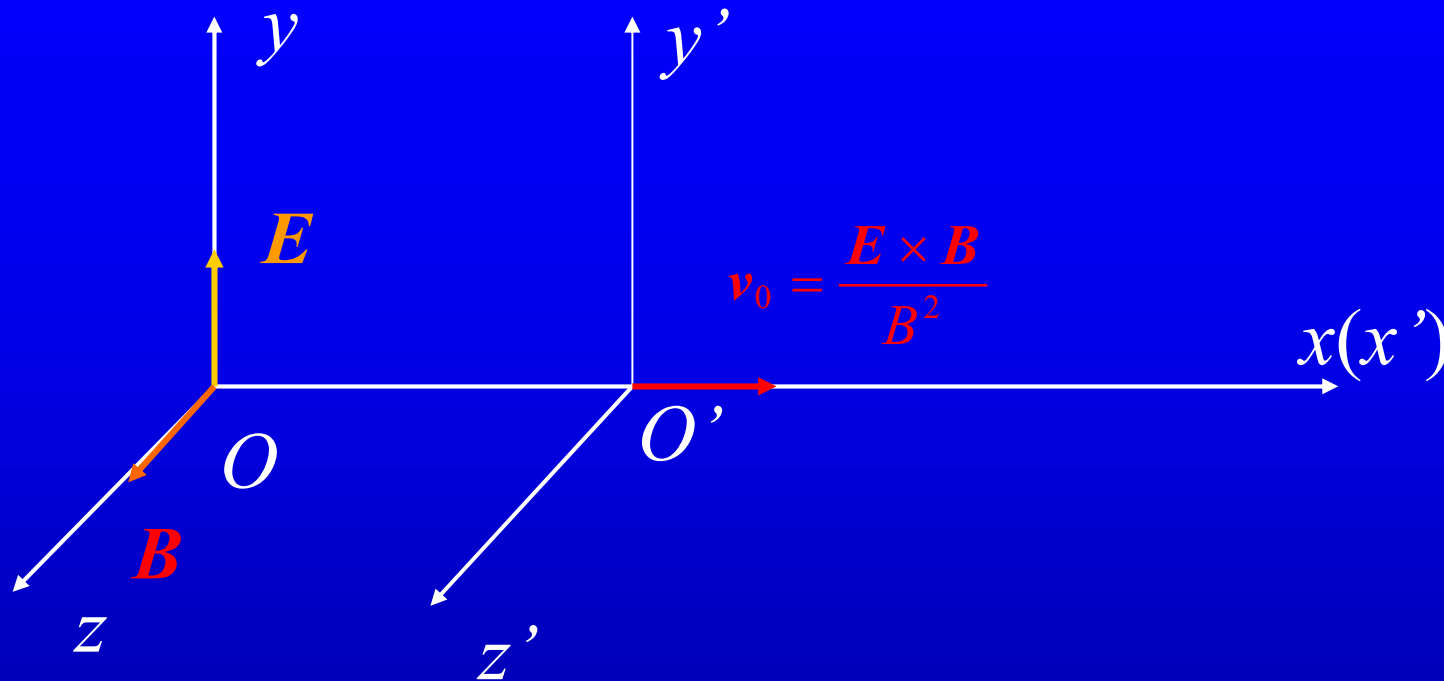
magnetic field -- point
into the photo

bubble chamber



Cloud chamber

*Motion of a charge in crossed electric and magnetic field



The equation of motion in the frame xyz is

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Make a frame $x'y'z'$ moving relative to xyz with velocity

$$\mathbf{v}_0 = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = \frac{E}{B} \mathbf{e}_x.$$

By a Galilean transformation from xyz to $x'y'z'$, the velocity of the particle in the frame $x'y'z'$ is $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$. Then, the equation of motion in the frame $x'y'z'$ is

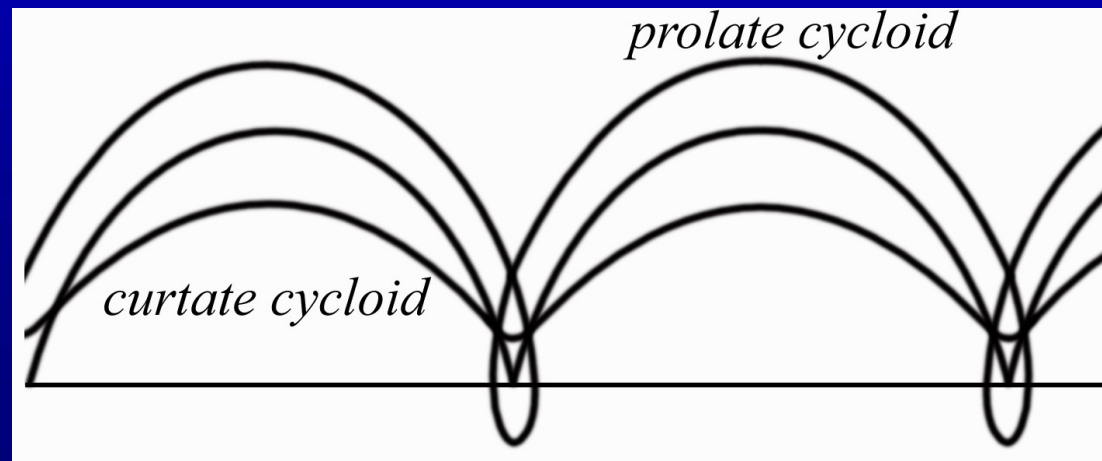
$$\begin{aligned} m \frac{d\mathbf{v}'}{dt} &= q(\mathbf{E} + \mathbf{v}' \times \mathbf{B} + \mathbf{v}_0 \times \mathbf{B}) \\ &= q(\mathbf{E} + \mathbf{v}' \times \mathbf{B} + \frac{E}{B} \mathbf{e}_x \times B \mathbf{e}_z) \\ &= q(\mathbf{E} + \mathbf{v}' \times \mathbf{B} + E(\mathbf{e}_x \times \mathbf{e}_z)) \\ &= q(\mathbf{E} + \mathbf{v}' \times \mathbf{B} - E \mathbf{e}_y) \\ &= q\mathbf{v}' \times \mathbf{B}. \end{aligned}$$

Note that the motion relative to $x'y'z'$ is as it would be if no electric field is present.

If the particle moves initially in xy -plane, the path of motion of the particle in $x'y'z'$ frame will be a circle

of radius $r = \frac{mv'}{qB}$ and the angular velocity is $\omega = -\frac{q}{m}\mathbf{B}$.

Then, relative to xyz , the circle advance with velocity v_0 . Therefore, if $\omega r = v' = v_0$ the path is the normal cycloid. But if $v' > v_0$ and $v' < v_0$, the paths are prolate and curtate cycloids respectively.



Applications

- (1) Bubble chamber.
- (2) Cathode-ray oscilloscope
- (3) Mass spectrometer
- (4) Cyclotron, synchrocyclotron and synchrotron.
- (5) Trapping of the charged particle

Wilson Cloud Chamber An apparatus that detects high-energy particles passing through a supersaturated vapor; each particle ionizes molecules along its path and small droplets condense on them to produce a visible track.

Bubble Chamber A particle detector of major importance during the initial years of high-energy physics. It is based on the principle of bubble formation a superheated liquid which is then suddenly expanded, starting boiling where passing charged particles have ionized the atoms of the liquid.

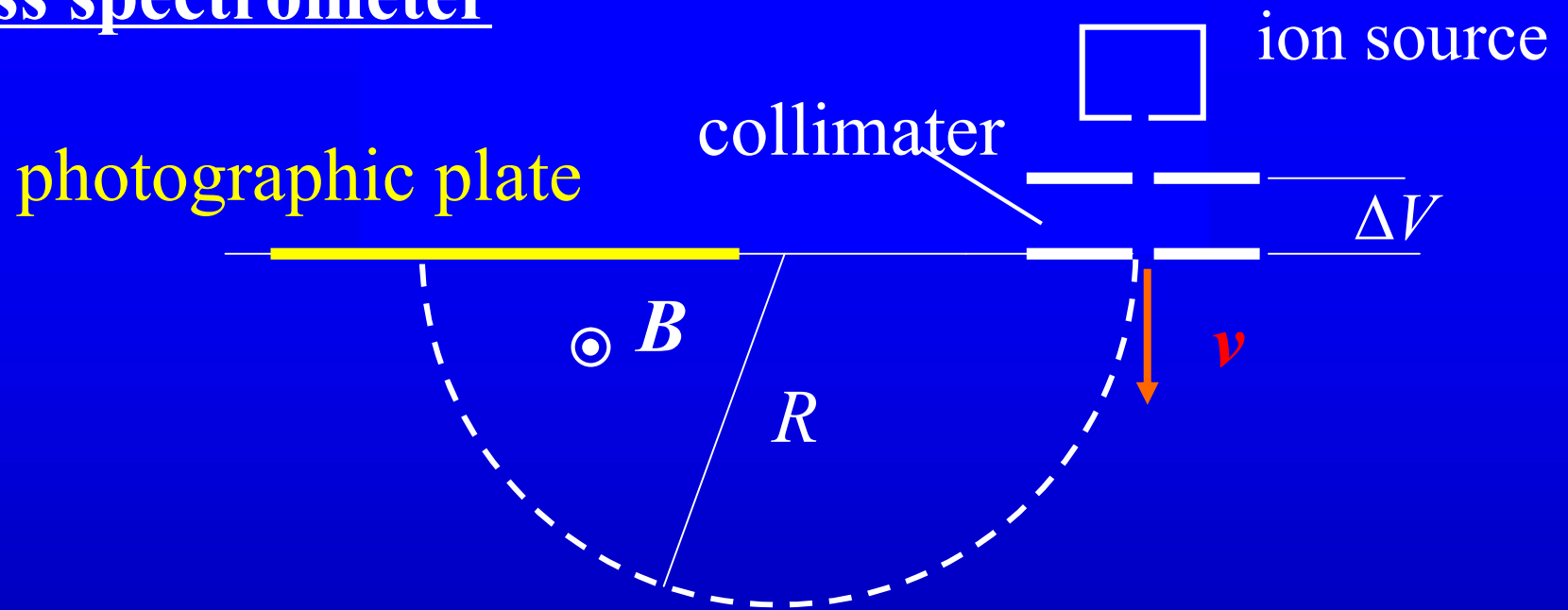
liquid hydrogen

propane (丙烷) C_3H_8

freons (氟利昂) Dupont's trade mark for fluor compounds, e.g. CF_2Cl_2 or CF_3Br).

This technique was honoured by the Nobel prize award to D. Glaser in 1960.

Mass spectrometer



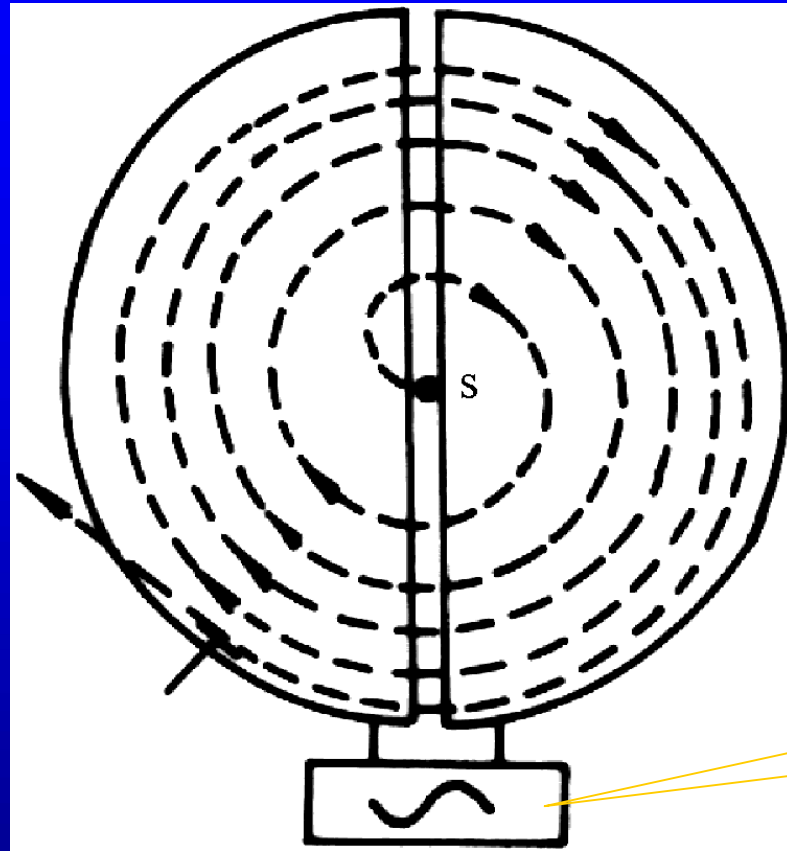
$$\frac{1}{2}mv^2 = q\Delta V.$$

$$v = \frac{qB}{m}R.$$



$$m = \frac{qB^2R^2}{2\Delta V}.$$

Cyclotron



$$\omega = \frac{qB}{m}$$

$$r = \frac{mv}{qB} = \frac{v}{\omega}$$

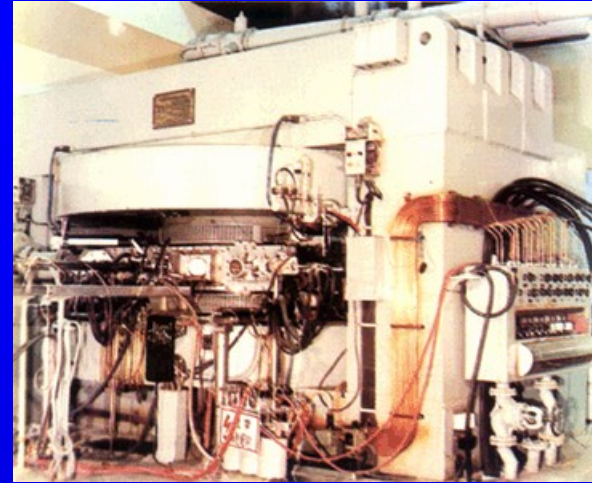
$$v_{\max} = \omega R = \frac{qBR}{m}$$

$$V_0 \sin \omega t$$

$$T_{\max} = \frac{1}{2} m v_{\max}^2 = \frac{q^2 B^2 R^2}{2m}$$



World's first cyclotron
made by E. O. Lawrence
in 1930. He won 1939
Nobel prize for physics.



China's first cyclotron
1958

When relativistic effect is taken into account, the
frequency of oscillator has to be adjusted

$$f_m = \frac{qB}{2\pi} = C \Rightarrow \text{Synchrocyclotron}$$

Synchrotron is a kind of “hollow cyclotron”.

R is fixed $\rho = \frac{v_{\perp}}{\omega} = \frac{mv_{\perp}}{qB}$ B has to be adjusted.

the frequency of oscillator also must be adjusted to match the variation of B and m.

Batavia accelerator

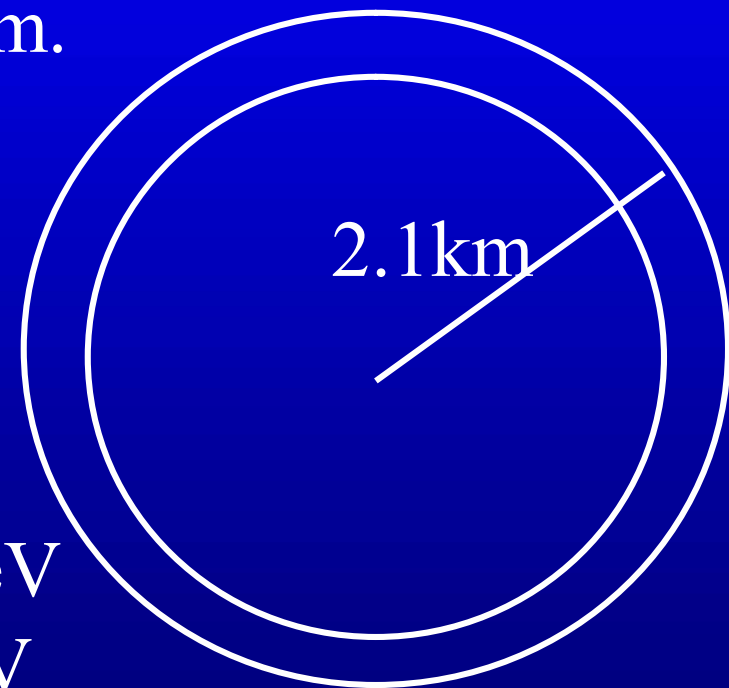
945 magnets $B_{\max} \sim 1.8\text{T}$

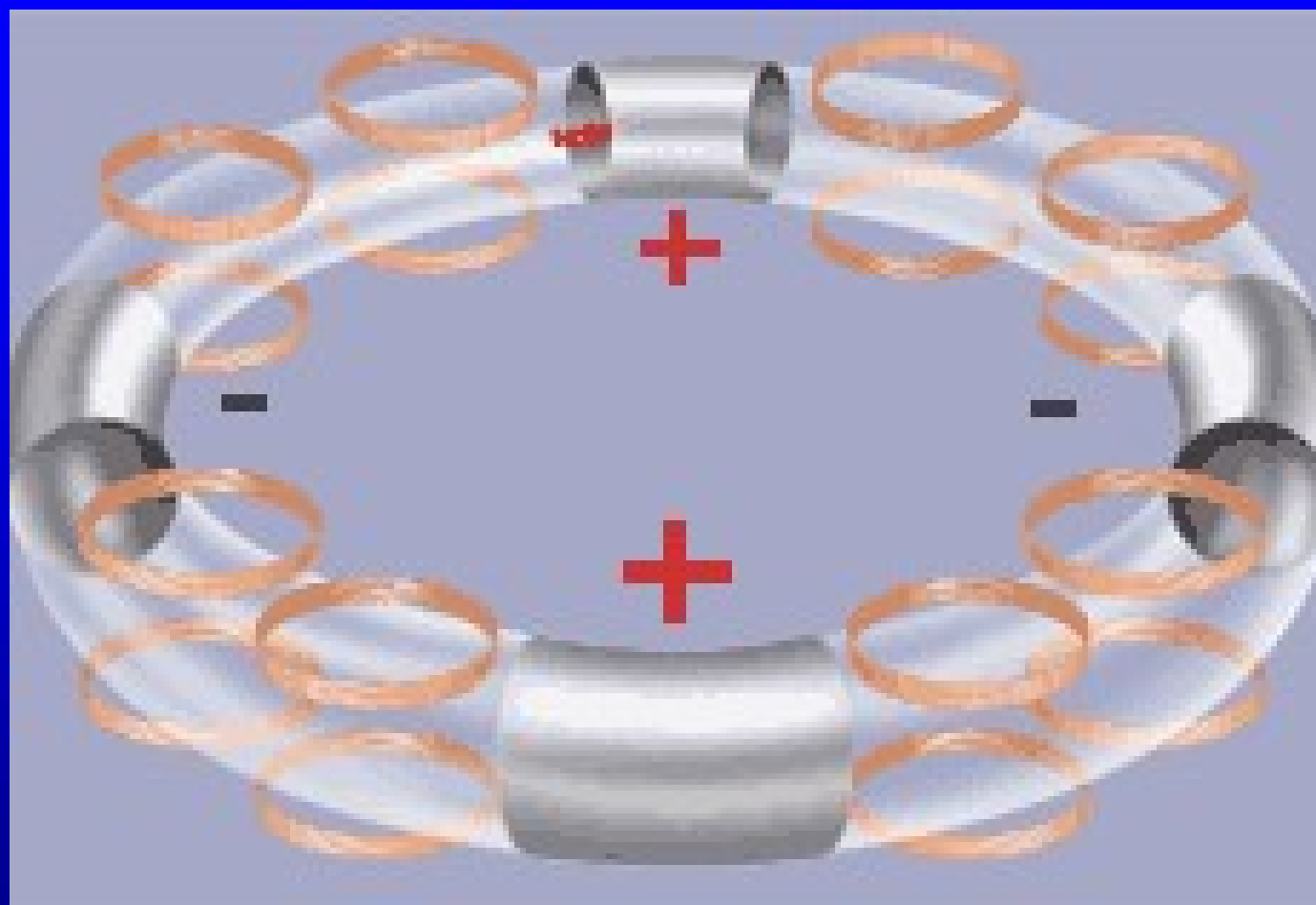
proton 500Gev

1MeV(megaelectronvolts)= 10^6eV

1GeV(gigaelectronvolts)= 10^9eV

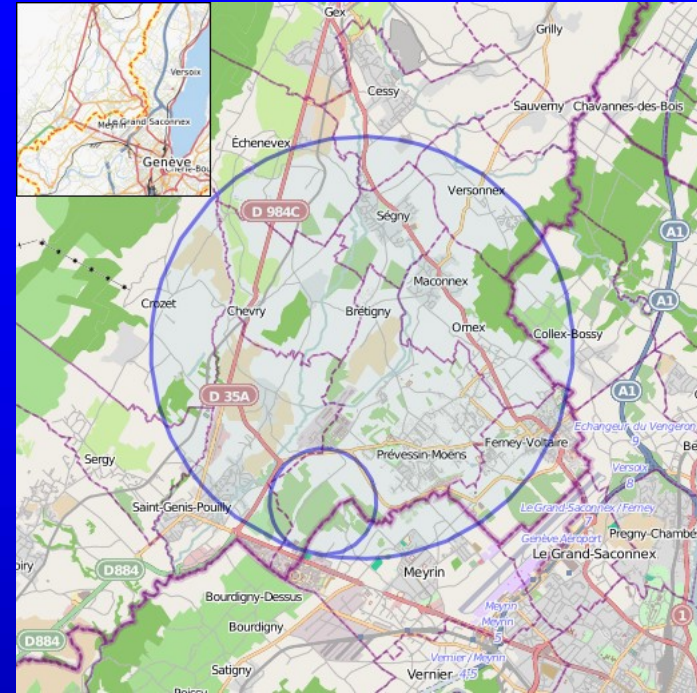
kilo mega giga







Tevatron of Fermi National Accelerator Laboratory (FNAL) 30 miles west of Chicago, the second highest energy particle collider in the world, inactive since 2011.



The Large Hadron Collider (LHC) is the world's largest and most powerful particle collider, most complex experimental facility ever built, and the largest single machine in the world.

circumference of 27 kilometres

6.5 TeV per beam (13 TeV total)

Magnetic mirror

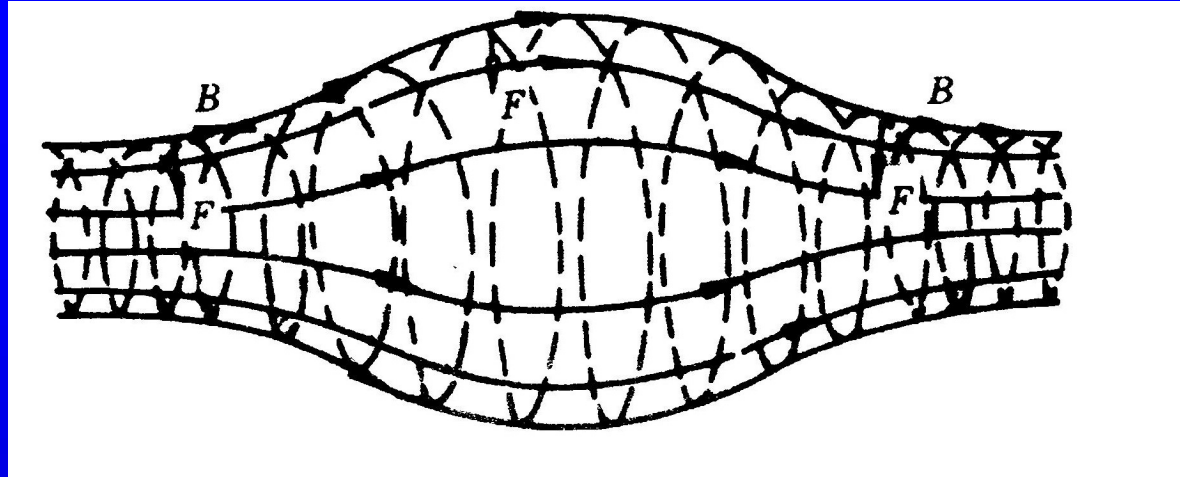
$$\frac{1}{2}mv^2 = \frac{1}{2}mv_{\perp}^2 + \frac{1}{2}mv_{\parallel}^2.$$

The kinetic energy of the particle is conserved when moving in a magnetic field

It can be proved that if the magnetic field is varied gradually in the space, The magnetic dipole moment of the particle is unchanged.

$$\mu = IS = qf\pi r^2 = q\left(\frac{\omega}{2\pi}\right)\pi r^2$$

$$= \frac{q\omega r^2}{2} = \frac{q}{2} \frac{qB}{m} \left(\frac{mv_{\perp}}{qB}\right)^2 = \frac{\frac{1}{2}mv_{\perp}^2}{B} = C.$$

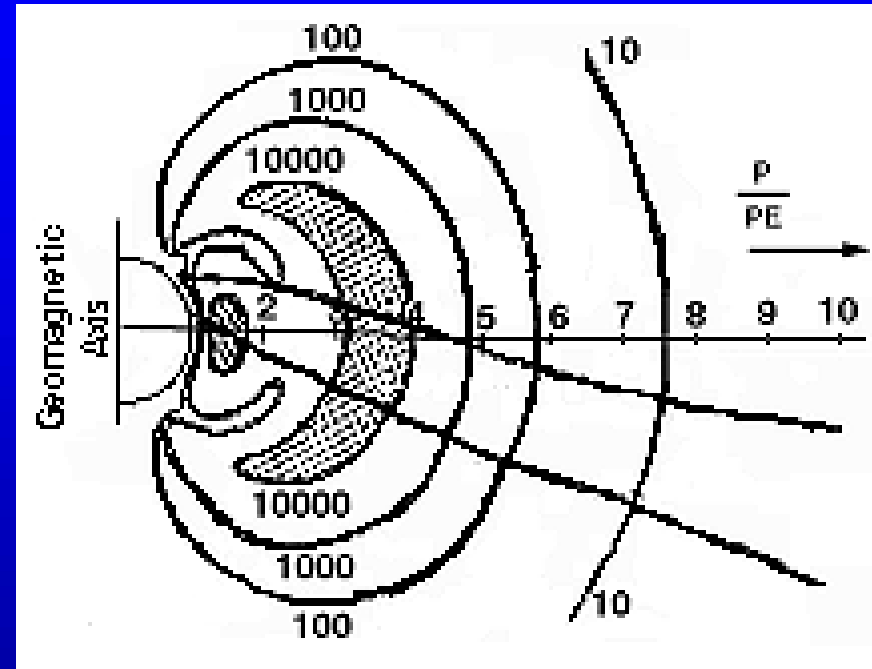
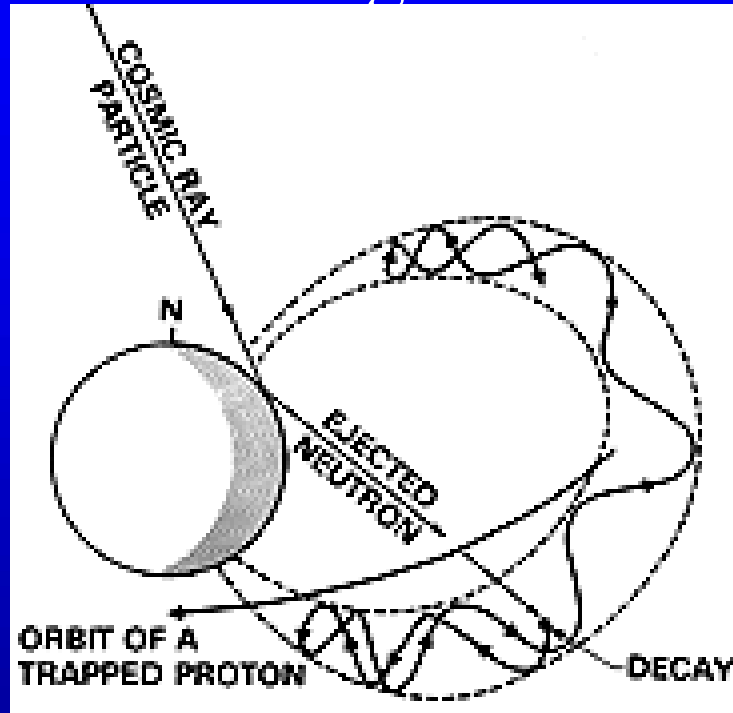


As a particle moves in the direction of increasing magnetic field, the parallel velocity reduces to zero and the particle is forced to move back.

magnetic bottle effect

TOKAMAK

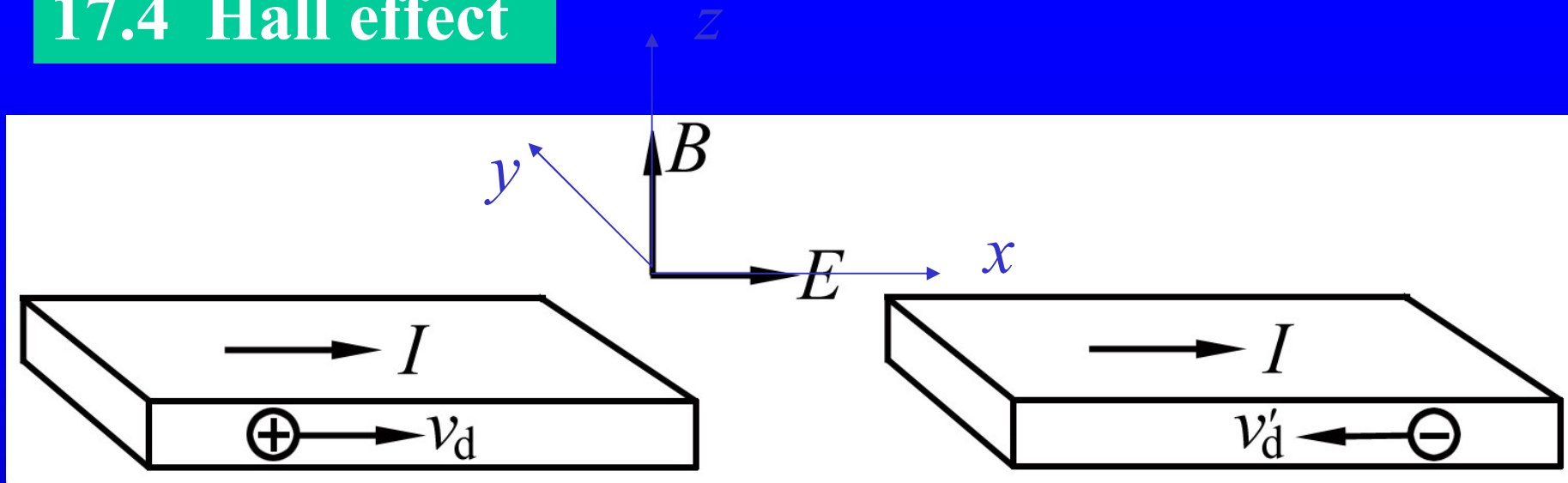
Van Allen radiation belts --rapidly moving charged particles, such as protons and electrons, trapped in earth's magnetic field.



inner belt----800-4000km

outer belt-----13000-19000km

17.4 Hall effect



The equilibrium requires that the force acted on the carrier produced by the transverse Hall electric field balances the force produced by the magnetic field. Hence, we have

$$q(\mathbf{E}_H + \mathbf{v}_d \times \mathbf{B}) = 0, \quad \mathbf{E}_H = -\mathbf{v}_d \times \mathbf{B}$$

1879(before 1897) , E.H.Hall

Substituting $\mathbf{v}_d = \frac{\mathbf{j}}{nq}$ into the above equation, the Hall field is

$$\begin{aligned}\mathbf{E}_H &= -\frac{1}{nq} \mathbf{j} \times \mathbf{B} = -\frac{1}{nq} j_x B \mathbf{e}_x \times \mathbf{e}_z \\ &= \frac{1}{nq} j_x B \mathbf{e}_y = E_y \mathbf{e}_y.\end{aligned}$$

Then, the Hall coefficient can be defined as

$$K = \frac{E_H}{j_x B} = \frac{1}{nq}.$$

$K > 0$	positive charge carrier
$K < 0$	negative charge carrier

Thus, the Hall effect provides a useful method to detect the sign of the charge carriers in conductors.

The normal conductivity is

$$\sigma = \sigma_{xx} = \frac{j_x}{E_x}.$$

The Hall conductivity can be defined as

$$\sigma_H = \sigma_{xy} = \frac{j_x}{E_y} = \frac{nq}{B}.$$

*Hall angle

$$\tan \theta_H = \frac{E_y}{E_x} = \frac{\frac{1}{nq} j_x B}{\frac{j_x}{\sigma}} = \frac{\sigma}{nq} B.$$

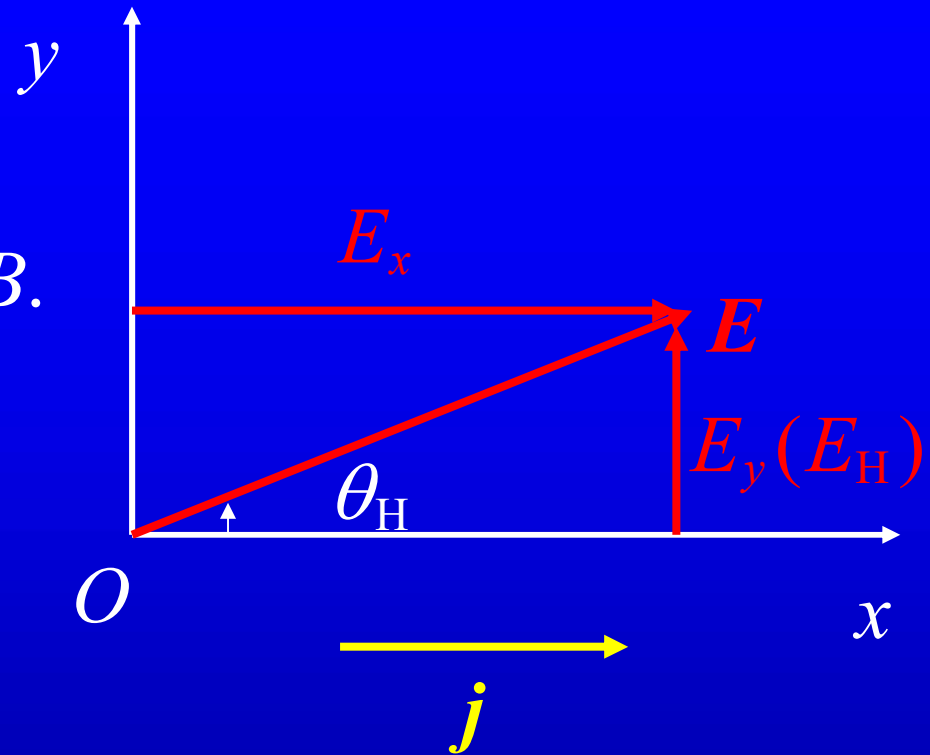
Substituting $\sigma = \frac{nq^2\tau}{m}$ into
above equation, we have

$$\tan \theta_H = \frac{qB}{m} \tau = \pm \omega \tau.$$

cyclotron frequency

+ (-)----positive(negative)
charge carrier

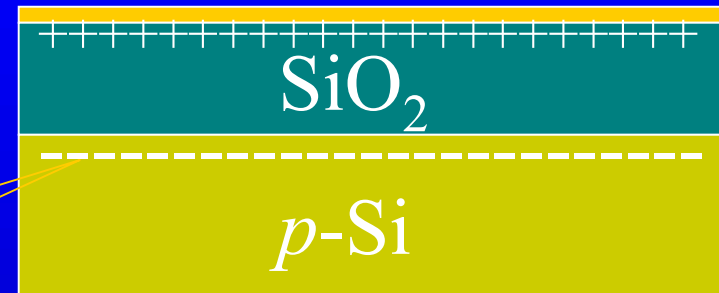
$$\omega \tau \ll 1 \Rightarrow \theta_H = \pm \omega \tau.$$



2D electron gas system

MOSFET

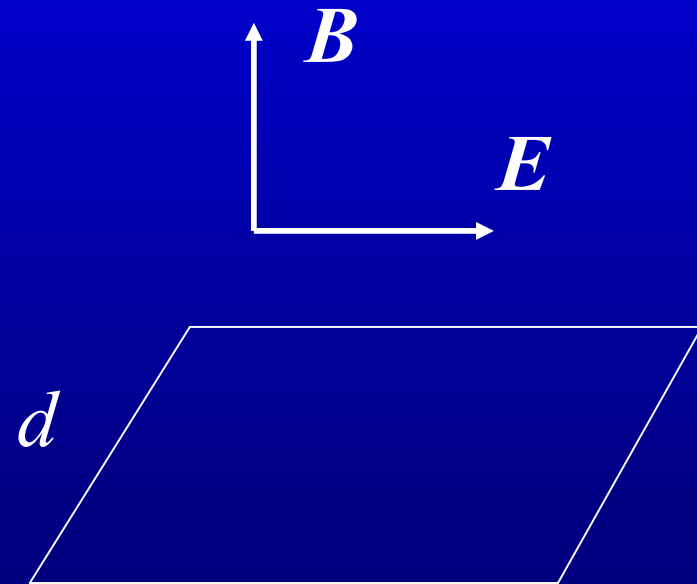
+V

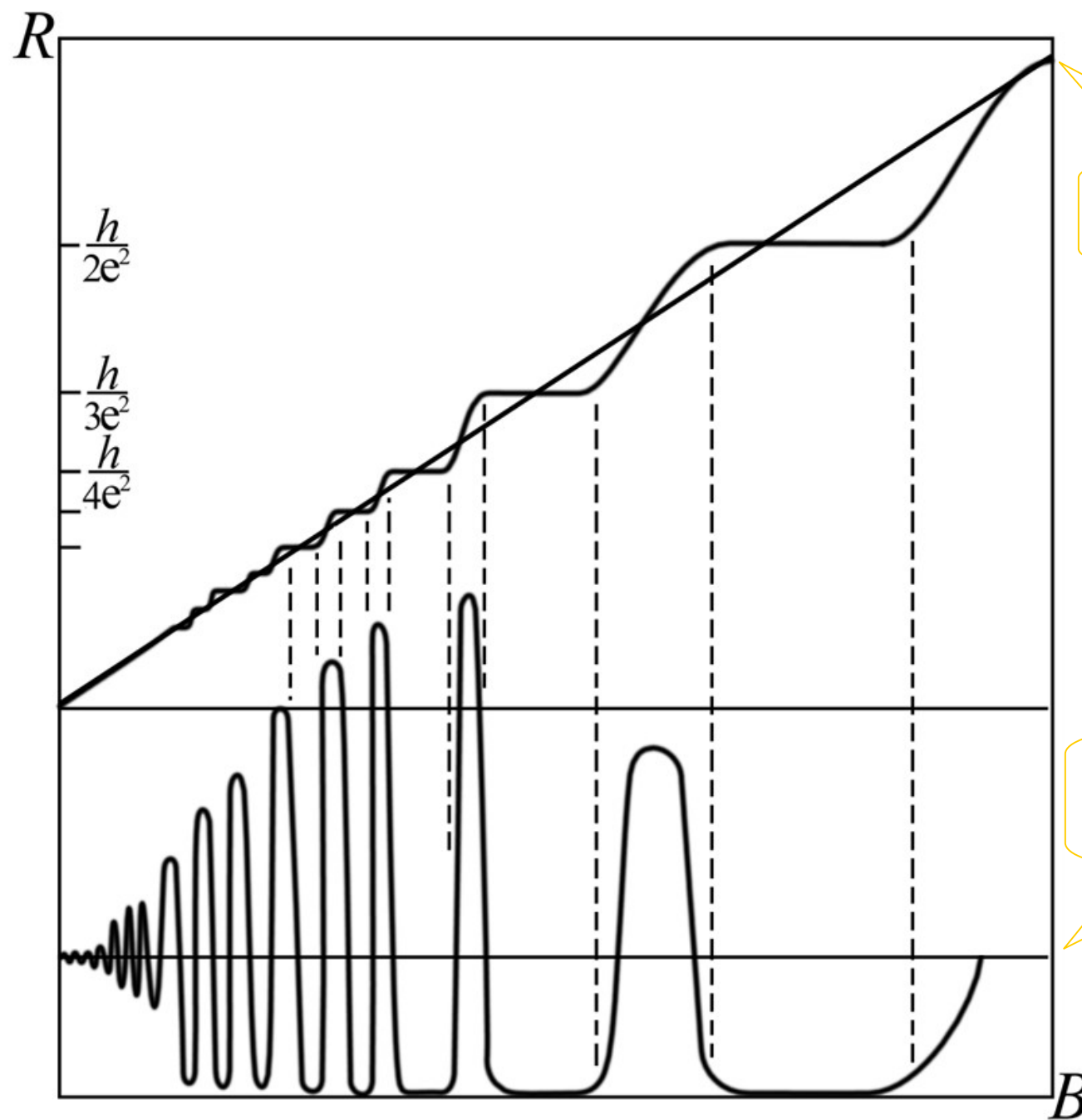


Inversion layer

In 2D system, Hall resistance is

$$R_H = \frac{V_H}{I} = \frac{V_H}{jd} = \frac{E_H}{j} = \frac{1}{\sigma_{xy}} = \rho_H$$





3D Hall R

3D normal

Quantum Hall effect (QHE)

$$R_H = \frac{h}{ne^2} \quad (n = 2, 3, \dots) \quad (1990)$$

$$h/e^2 = 25812.806\Omega$$

same for different geometries and processing histories, and even for those made of a variety of materials

It can be used to determine fine structure constant.

$$\alpha = \frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} = \frac{1}{137.036}$$

In 1980, Klaus von Klitzing found the integral quantum Hall effect (IQHE) and later won the 1985 Nobel Prize in physics.

In 1982, D. C. Tsui et al. found the fractional (FQHE) quantum Hall effect.

(GaAs $\text{Al}_x\text{Ga}_{1-x}\text{As}$ heterostructure)

1998 Nobel laureates: H. L. Stormer , D. C. Tsui,
R. B. Laughlin.

IQHE---- Landau level+impurities

FQHE----incompressible quantum liquid.

Laughlin wave function

quantum Hall effect ~
high temperature superconductivity

Low dimensional system

Strongly correlated electron system

Assignment: 17.9, 17.11